



v7.2.0

Zkong Networks Electronic Shelf Label System API Documentation

Global Leader in Cloud ESL
WWW.ZKONG.COM

Document Revision History

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1	v1.0	2020/07/30	Hu Zhaoxu	Zkong ESL System API Documentation v1.0 compiled
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1. Overview

This document is intended for users who need to integrate their ERP system with the Zkong ESL system. It explains how to connect data between the ERP platform and the ESL platform, allowing users to manage daily operations in the ERP system while the ESL system updates automatically.

Notes:

1. All business APIs require authentication. Please obtain a token through the login API and add it to the request header for authentication: Authorization: token.
2. API status-code messages support multilingual responses. To specify the response language, add the Language parameter to the request header. The default value is Language: cn, and the value is not case-sensitive. Supported languages include Simplified Chinese (cn), Traditional Chinese (tw), English (en), Polish (pl), Russian (ru), Japanese (ja), Arabic (ar), Spanish (es), Korean (ko), Ukrainian (ua), Thai (tha), and Hebrew (he).
3. APIs that support an external store ID can be called using that external store ID. The external store ID must be configured in the ESL system in advance and is always passed as a query parameter.

Example: Using an external store ID for batch product import:

<http://esl.zkong.com/zk/item/batchImportItem?useExternalStoreId=1&externalStoreId=1011201>

2. Get the Public Key and Log In to ESL

2.1 Get RSA Public Key

API Definition

URL	http://[server address]/zk/user/getErpPublicKey
Example	http://esl.zkong.com/zk/user/getErpPublicKey
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Get the RSA public key used to encrypt the account password for the login API, log in to the ESL platform, and obtain user information and a valid token. (This public key may change. If login fails, obtain it again.) See Appendix 2 for an encryption example.

Request Parameters

None

Request Example

Request URL:
<http://esl.zkong.com/zk/user/getErpPublicKey>
Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false

code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Operation successful/failed
data	Response Data	String	Public key string

Response Example (related error code no.: 1)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": "MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCRxoFbdbaFhTFVG7iAXFVy
    L21c6y/Bip5Zd57RAw3vER0XewcpuzULj2DCxJmbTCjRvhG84qLVMv1qL7RP
    AHT48uZ5bRt36gin/Lj80mD3FPPXoPnI0ZC+48zHHz/phM4nb3mvrrD4Uhsi
    eDi72rV8/K4z4kA736dnnga7X0D1wIDAQAB"
}
```

2.2 Log In to ESL Platform

API Definition

URL	http://[server address]/zk/user/login
Example	http://esl.zkong.com/zk/user/login
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Use the account and password to log in to the ESL platform and obtain user information and a valid token (valid for 7 days). See Appendix 2 for an example of account/password encryption. (If login fails, call API 2.1 again to obtain the public key and retry.) (If a business API returns status code 401, log in again to obtain a new token.)

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Description
account	body	String (100)	true	Login account
password	body	String	true	Password (must be

				encrypted using the RSA public key)
loginType	body	Integer (1)	true	Fixed value: 3 (ERP login)

Request Example

Request URL:

<http://esl.zkong.com/zk/user/login>

Request Body:

```
{
  "account": "zkongtest",
  "loginType": 3,
  "password": "bsYH2jBN+jbc+dX1EAZm9EKjPojpCF+3iPvN3k3lKdrJjf2XNpZdWZ+Dn
+QCDzopFZiK685NbI3NiwrV/M10eeCuPhEQ3XQUiOPsJnh403jtHb3w
HoUvqjy+LvGU2s3mJU/aoPhOWASdng77tdGvaJcjf6xXHQTxcfw/C4L
Yx+8="
}
```

Response Parameters

Parameter Name		Description	Data Type	Notes
success		Request Successful	Boolean	true/false
code		Status Code	Integer	See Appendix 1 for details
message		Message	String	
data		Response Data	Json	
	roleCode	Role Code	Integer	
	userName	Username	String	
	token	Token	String	Used for business API authentication
	currentUser	User Information	Json	Information of the currently logged-in user
	id	User ID	Integer	

	account	Account Name	String	
	password	Password	String	
	roleId	Role ID	Integer	
	agencyId	Agency ID	Integer	
	agencyName	Agency Name	String	
	merchantId	Merchant ID	Integer	
	merchantName	Merchant Name	String	
	storeId	Store ID	Integer	
	storeName	Store Name	String	
	enable	Enabled	Integer	0: disabled, 1: enabled
	flag	Valid	Integer	0: invalid, 1: valid
	name	Name	String	
	phone	Phone	String	
	email	Email	String	
	createTime	Created Time	Date	
	lastLoginTime	Last Login Time	Date	

Response Example (related error code no.: 1, 2, 3, 4, 5, 6, 7, 8, 9)

```
{
  "success": true,
  "code": 14014,
  "message": "Login successful",
  "data": {
    "currentUser": {
      "id": 46,
      "account": "abc123",
      "password": "21232f297a57a5a743894a0e4a801fc3",
      "roleId": 1,
      "createTime": "2019-03-27 17:24:09",
      "lastLoginTime": "2019-07-04 14:11:51",
      "token": null,
      "merchantId": "",
      "merchantName": null,
      "storeId": null,
      "storeName": null,
      "enable": 1,
      "flag": 1,
      "name": null,
      "phone": null,
      "email": null,
      "department": null,
      "agencyId": null,
      "agencyName": null
    },
    "roleCode": "admin",
    "userName": "admin",
    "token": "eyJhbGciOiJIUzI1NiJ9.eyJleHAiOjE1OTQ5NjAxMTIsInRpbWVzdGFtcCI6MTU5NDk1NjUxMjUzNH0.z3-DbOG0AZtwev0_Vn_0sIr1NU7symTjXkqe0VaJR0o"
  }
}
```

Example 2

```
{
  "success": false,
  "code": 14018,
  "data": null,
  "message": "Username does not exist",
  "translate": true,
  "originalMessage": null
}
```

3. Product Information Synchronization and Query

3.1 Batch Import Products

API Definition

URL	http://[server address]/zk/item/batchImportItem
Example	http://esl.zkong.com/zk/item/batchImportItem
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Import product data into the ESL platform through the HTTP API. 1) When a store imports new products for the first time, the basic product information is synchronized under the merchant. Other stores that have data synchronization enabled can also use this product data. 2) When importing products through the API, if storeId is empty, the products are imported into the overall product list. If storeId is not empty, the product information is imported only into the corresponding store. 3) External store ID is supported for product import. The external store ID must be configured in the ESL system in advance.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer (1)	false	Whether to use externalStoreId (0: no, 1: yes)
externalStoreId	query	String (100)	false	External store ID (required when used)
emptyNeedDelete	query	Integer (1)	false	Overwrite rule when product attributes are updated to empty

				values: 0 keep the original value, 1 overwrite with an empty value (default: 1)
requestId	body	String(255)	false	Request ID
storeId	body	Integer (20)	false	Store ID
merchantId	body	Integer (20)	true	Merchant ID
agencyId	body	Integer (20)	true	Agency ID
unitName	body	Integer (1)	false	Price unit, default: 0 0: Only integers are allowed, length 0-16. Price fields are divided by 100 and kept to two decimal places. 1: Supports 14 integer digits and 5 decimal places
timezone	body	String(100)	false	Time zone GMT+08:00/GMT-05:00 (optional; used for automatic price change time conversion)
repricingFullUpdate	body	Integer	false	Whether

					promotion activities are fully updated. 1: full update (clear previous promotion activities for this product and then add new ones); 0: incremental update (default, appended by approval time - creation time - collection order)
	itemList	body	List<Item> (20000)	true	Product list (single request limit: 20,000 records)
	barCode	body	String (100)	true	Product Code (unique product identifier)
	attrCategory	body	String (100)	true	Template Category
	attrName	body	String (100)	true	Template Attribute
	productCode	body	String (1000)	false	Product barcode. Separate multiple values with commas
	productSku	body	String (100)	false	Product SKU
	itemTitle	body	String	false	Product Title

		(500)		
shortTitle	body	String (500)	false	Short Product Name
classLevel	body	String (100)	false	Grade
productArea	body	String (100)	false	Place of Origin
unit	body	String (100)	false	Sales Unit
qrCode	body	String (1000)	false	QR Code URL
nfcUrl	body	String (1000)	false	NFC Link
spec	body	String (100)	false	Specification
originalPrice	body	BigDecimal [20,5]	false	Original price (negative values are not supported)
price	body	BigDecimal [20,5]	false	Sales price (negative values are not supported)
memberPrice	body	BigDecimal [20,5]	false	Member price (negative values are not supported)
stock1	body	BigDecimal	false	Stock 1

			[10,2]		(negative values are supported)
stock2	body	BigDecimal	false	[10,2]	Stock 2 (negative values are supported)
stock3	body	BigDecimal	false	[10,2]	Stock 3 (negative values are supported)
proStartTime	body	String	false		yyyy-MM-dd HH:mm:ss format
proEndTime	body	String	false		yyyy-MM-dd HH:mm:ss format
promotionText	body	String	false	(100)	Promotion Text
custFeature1	body	String	false	(1000)	Custom Extended Field 1
custFeature2	body	String	false	(1000)	Custom Extended Field 2
...	...	...	...	...	...
custFeature50	body	String	false	(100)	Custom Extended Field 50
label	body	String	false	(20)	Label
measurementType	body	Integer	false		Measurement type: 1 - by weight, 2 - by

					count
	listingStatus	body	Integer	false	Listing status: 1 - listed, 2 - delisted
	labelNum	body	String (100)	false	Label Number
	primaryCategory	body	String (100)	false	Primary Product Category
	secondaryCategory	body	String (100)	false	Secondary Product Category
	repricingList	body	List<Item> (30)	false	Automatic Price Change List (maximum: 30)
	repricingType	body	Integer	true	Promotion order type, determines which configuration field group to map to (1 - Price 1, 2 - Price 2, 3 - Price 3)
	auditTime	body	String (50)	false	Approval Time
	repricingDateStart	body	String (50)	true	Promotion start date. Supported formats: yyyy-MM-dd, yyyy/MM/dd, yyyy-MM-dd in Chinese date

						format, and date ranges separated by ~
		repricingDateEnd	body	String (50)	false	Promotion end date, same format as above
		repricingTimeStart	body	String (50)	true	Promotion start time. Supported formats: HH:mm or HH:mm in Chinese time format
		repricingTimeEnd	body	String (50)	true	Promotion end time, same format as above
		price	body	BigDecimal [20,5]		Promotion price, either this field or discount is required
		discount	body	BigDecimal [10,2]		Discount, either this field or price is required
		discountCalculationField	body	String (100)		Discount calculation field required when discount is provided
		copyText	body	String (50)	false	Text

Note: extended fields 1-3 have a length limit of 1000, fields 4-10 have a limit of 500, fields 11-15 have a limit of 200, and fields 16-50 have a limit of 100.

Request Example

Request URL:

<http://esl.zkong.com/zk/item/batchImportItem?useExternalStoreId=1&externalStoreId=1011201>

Request Body:

```
{
  "agencyId": "1629967488587",
  "merchantId": "1629968747662",
  "storeId": "1630484444737",
  "unitName": 1,
  "itemList": [
    {
      "attrCategory": "default",
      "attrName": "default",
      "barCode": "100001234",
      "itemTitle": "Red Fuji Apple",
      "originalPrice": "600",
      "price": "6300",
      "productArea": "Yantai, Shandong",
      "productCode": "200001582",
      "productSku": "200001275",
      "promotionText": "Promotion Text",
      "custFeature1": "Buy five, get one free"
    },
    {
      "attrCategory": "default",
      "attrName": "default",
      "barCode": "10004562",
      "itemTitle": "Hainan Banana",
      "originalPrice": "10",
      "price": "8",
      "productArea": "Hainan, China",
      "productCode": "200001423",
      "productSku": "20001536",
      "promotionText": "Promotion Text"
    }
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Json	Product error information see Example 2

Response Example (related error code no.: 3, 10~23)

```
{
  "success": true,
  "code": 17002,
  "data": null,
  "message": "successful import of merchandise"
}
```

Example 2

```
{
  "success": false,
  "code": 17051,
  "message": null,
  "data": {
    "temp_attr_none": [
      "10004562"
    ],
    "erp_import_original_price_invalid": [
      "100001234"
    ]
  },
  "translate": true,
  "originalMessage": null
}
```

3.2 Batch Delete Products

API Definition

URL	http://[server address]/zk/item/batchDeleteItem
Example	http://esl.zkong.com/zk/item/batchDeleteItem
Request Method	DELETE
Content Type	application/json;charset=utf-8
Description	Delete products from the ESL platform through the HTTP API. 1) If storeId is empty, the product information with the corresponding barcode will be deleted from all stores under the merchant. 2) A maximum of 500 products can be deleted at a time. 3) External store ID is supported for product deletion. The external store ID must be configured in the ESL system in advance.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer(1)	false	Whether to use externalStoreId: 0 do not use, 1 use
externalStoreId	query	String (40)	false	External Store ID required when used
storeId	body	Integer(20)	false	Store ID
list	body	List<String> (500)	true	Product Code List

Request Example

Request URL:
`http://esl.zkong.com/zk/item/batchDeleteItem`

Request Body:

```
{
  "storeId": "1588041156676",
  "list": [
    "2000123456",
    "2000135798"
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 10, 12, 24, 25, 26, 27)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 19001,
  "message": "Product list is empty",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

3.3 Upload Image Product

API Definition

URL	http://[server address]/zk/item/uploadPicItem
Example	http://esl.zkong.com/zk/item/uploadPicItem
Request Method	POST
Content Type	multipart/form-data
Description	Upload an image product to the ESL platform through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
----------------	----------------	-----------	----------	-------

Authorization	header	String	true	Authorization token
merchantId	query	Integer(20)	true	Merchant ID
barCode	query	String(100)	true	Product Code
modelColor	query	String(1)	true	ESL Supported Colors 0: black/white, 1: black/white/red 2: black/white/yellow, 4: seven-color type 1 5: black/white/red/yellow, 6: seven-color type 2 7: full color
modelSize	query	String(20)	true	ESL Size, e.g. 2.13
picSize	query	String(20)	true	Image size, e.g. 250*122
itemTitle	query	String(100)	false	image productname
parseAlgorithm	query	Integer (1)	false	Image parsing method: 0 Floyd-Steinberg dithering, 1 normal indexing
storeId	query	Integer(20)	false	Store ID
nfcUrl	query	String	false	NFC Link
multipartFile	body	File(600KB)	true	product image limit600KB

Request Example

Request URL:
http://esl.zkong.com/zk/item/uploadPicItem?merchantId=1585812365204&barcode=2000123456&modelColor=1&modelSize=2.13&picSize=250*122
Request Body:

```
{
  "multipartFile": "<uploadfile>"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request	Boolean	true/false

	Successful		
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 3 16 18 25 28 29 30 31 32 33 34 35 36 37 38)

```
{
  "success": true,
  "code": 17001,
  "message": "Submitted successfully",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 10038,
  "message": "Image size is too large",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

3.4 Upload Product Image Information

API Definition

URL	http://[server address]/zk/itemPic/uploadItemPic
Example	http://esl.zkong.com/zk/itemPic/uploadItemPic
Request Method	POST
Content Type	multipart/form-data
Description	Upload product images to the ESL platform through the HTTP API. This API is used to upload image information for ordinary products. Note the difference from the image product API in section 3.3.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
merchantId	query	Integer (20)	true	Merchant ID
storeId	query	Integer (20)	false	Store ID (required when uploading to a store)
barCode	query	String (100)	true	Product Code
picNum	query	Integer (2)	true	image number (1-10)
multipartFile	body	File (600KB)	true	product image(sizelimit600KB)

Request Example

Request URL:
<http://esl.zkong.com/zk/item/uploadPicItem?merchantId=1585812365204&storeId=1585812365275&barCode=2000123456&picNum=1>

Request Body:

```
{
  "multipartFile": "<uploadfile>"
}
```

Response Parameters

Parameter Name	description name	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 16 25 27 28 30 31 39 40)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 17004,
  "message": "Product does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

3.5 Delete Product Image

API Definition

URL	http://[server address]/zk/item/delItemPic
Example	http://esl.zkong.com/zk/item/delItemPic
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Delete product images from the ESL platform through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

barcode	body	String(100)	true	Product Code
picIndex	body	List<Integer>(10)	true	Image number(s) to delete for this product image numberlist (1-10)
storeId	body	Integer(20)	false	storeId (if not provided, merchant-level data is deleted)

Request Example

Request URL:
<http://es1.zkong.com/zk/item/delItemPic>
 RequestBody:

```
{
  "barcode": "1580114",
  "picIndex": [
    1,
    2
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 41)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 10009,
  "message": "Parameter error",
  "data": null,
  "translate": true,
  "originalMessage": "param invalid"
}
```

3.6 Get Product Information by Product Code

API Definition

URL	http://[server address]/zk/item/getItemByBarCodeAndExternalStoreId
Example	http://esl.zkong.com/zk/item/getItemByBarCodeAndExternalStoreId
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Get product information by product code. When externalStoreId is empty, merchant-level data is queried; when it is not empty, store-level data is queried.

Request Parameters

Parameter Name	Parameter Type	Parameter Type	Required	Notes
Authorization	header	String	true	Authorization token
itemBarCode	body	String(100)	true	Product Code
externalStoreId	body	String(40)	false	External Store ID

Request Example 1

Request URL:

<http://esl.zkong.com/zk/item/getItemByBarCodeAndExternalStoreId>

RequestBody:

```
{
  "itemBarCode": "123456",
  "externalStoreId": "10001"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 13 27)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "id": 11174894361,
    "type": 1,
    "price": null,
    "spec": "",
    "productCode": "",
    "barCode": "100001234",
    "agencyId": 1629967488587,
    "merchantId": 1629968747662,
    "storeId": 0,
    "originalPrice": null,
    "memberPrice": null,
    "productArea": "",
    "classLevel": "",
    "attrCategory": "default",
    "attrName": "default",
    "sourceCode": "",
    "proStartTime": null,
    "proEndTime": null,
    "unit": null,
    "productSku": "",
    "createdTime": "2023-02-27 19:28:44",
  }
}
```

```
        "itemTitle": "Red Fuji Apple",
        "shortTitle": "",
        "qrCode": "",
        "promotionText": "",
        "custFeature1": "",
        "custFeature2": "",
        "custFeature3": "",
        "custFeature4": "",
        "custFeature5": "",
        "custFeature6": "",
        "custFeature7": "",
        "custFeature8": "",
        "custFeature9": "",
        "custFeature10": "",
        "custFeature11": "",
        "custFeature12": "",
        "custFeature13": "",
        "custFeature14": "",
        "custFeature15": "",
        "custFeature16": "",
        "custFeature17": "",
        "custFeature18": "",
        "custFeature19": "",
        "custFeature20": "",
        "custFeature21": "",
        "custFeature22": "",
        "custFeature23": "",
        "custFeature24": "",
        "custFeature25": "",
        "custFeature26": "",
        "custFeature27": "",
        "custFeature28": "",
        "custFeature29": "",
        "custFeature30": "",
        "custFeature31": "",
        "custFeature32": "",
        "custFeature33": "",
        "custFeature34": "",
        "custFeature35": "",
        "custFeature36": "",
        "custFeature37": "",
        "custFeature38": "",
        "custFeature39": "",
        "custFeature40": "",
        "custFeature41": "",
        "custFeature42": "",
        "custFeature43": "",
        "custFeature44": "",
        "custFeature45": "",
        "custFeature46": "",
        "custFeature47": "",
        "custFeature48": "",
        "custFeature49": "",
        "custFeature50": "",
        "updateTime": "2023-05-25 11:57:10",
        "stock1": null,
        "stock2": null,
        "stock3": null,
        "nfcUrl": "",
        "storeName": null,
        "label": "",
        "number": null,
        "brandName": null,
        "suggestPrice": null,
        "coverPic": null,
        "delTime": null,
        "isSpecs": 0
    },
    "translate": true,
    "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 19015,
  "message": "No store matches the external store ID",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

3.7 Get Product Information Bound to ESLs Under a Merchant

API Definition

URL	http://[server address]/zk/bind/geyltemByPriceTagBarCode
Example	http://esl.zkong.com/zk/bind/geyltemByPriceTagBarCode
Request Method	POST
Content Type	application/json;charset=utf-8

Description	Get product information bound to ESLs under the same merchant.
--------------------	--

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
priceTagBarCode	body	String(60)	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/geyItemByPriceTagBarCode>
 Request Body:

```
{
  "priceTagBarCode": "100058526"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	Product Information

Response Example (related error code no.: 42 43)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "id": 1000418055,
    "type": 1,
    "price": 3,
    "spec": "",
    "productCode": "8521",
    "barCode": "123456789",
  }
}
```

```
"agencyId": 1555414082383,
"merchantId": 1615365123378,
"storeId": 1615365123392,
"originalPrice": 3.35,
"memberPrice": 2.35,
"productArea": "China",
"classLevel": "level",
"attrCategory": "default",
"attrName": "default",
"sourceCode": "",
"proStartTime": "2018-10-10 10:00:00",
"proEndTime": "2018-10-10 10:00:00",
"number": null,
"unit": "",
"productSku": "123",
"createdTime": "2021-09-07 09:44:20",
"itemTitle": "Red Fuji Apple",
"shortTitle": "cola",
"brandName": null,
"suggestPrice": null,
"qrCode": "",
"promotionText": "discount",
"coverPic": null,
"custFeature1": "",
"custFeature2": "",
"custFeature3": "",
"custFeature4": "",
"custFeature5": "",
"custFeature6": "",
"custFeature7": "",
"custFeature8": "",
"custFeature9": "",
"custFeature10": "",
"custFeature11": "",
"custFeature12": "",
"custFeature13": "",
"custFeature14": "",
"custFeature15": "",
"custFeature16": "",
"custFeature17": "",
"custFeature18": "",
"custFeature19": "",
"custFeature20": "",
"custFeature21": "",
"custFeature22": "",
"custFeature23": "",
"custFeature24": "",
"custFeature25": "",
"custFeature26": "",
"custFeature27": "",
"custFeature28": "",
```

```
"custFeature29": "",
"custFeature30": "",
"custFeature31": "",
"custFeature32": "",
"custFeature33": "",
"custFeature34": "",
"custFeature35": "",
"custFeature36": "",
"custFeature37": "",
"custFeature38": "",
"custFeature39": "",
"custFeature40": "",
"custFeature41": "",
"custFeature42": "",
"custFeature43": "",
"custFeature44": "",
"custFeature45": "",
"custFeature46": "",
"custFeature47": "",
"custFeature48": "",
"custFeature49": "",
"custFeature50": "",
"updateTime": null,
"delTime": null,
"stock1": null,
"stock2": null,
"stock3": null,
"isSpecs": 0,
"nfcUrl": "",
"storeName": null
}
}
```

Example 2

```
{
  "success": false,
  "code": 13040,
  "message": "Binding relationship does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

3.8 Query Store Product Information by Page

API Definition

URL	http://[server address]/zk/erp/item/list
-----	--

Example	http://esl.zkong.com/zk/erp/item/list
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query store product information by condition with pagination.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Token value returned by the login API
page	query	Integer	false	Page Number Default: 1(page 1)
size	query	Integer	false	Page Size Default: 10
useExternalStoreId	query	Integer(1)	false	Use externalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	externalStoreId, required when used
storeId	body	Integer(20)	false	Store ID. If not provided, returns merchant-level product information. It can be replaced by external store ID.
attrClass	body	String(100)	false	Template Category
attrName	body	String(100)	false	Template Attribute
pcltemTitle	body	String(60)	false	Product Name
pcBarCode	body	String(100)	false	Product Code
label	body	String(32)	false	Product Label

pcProductCode	body	String(1000)	false	Product Barcode
---------------	------	------------------	-------	-----------------

Request Example 1

Request URL:

<http://esl.zkong.com/zk/erp/item/list?page=1&size=10>

Request Body:

```
{
  "storeId": 1571661761656,
  "attrClass": "default",
  "attrName": "default"
}
```

Request Example 2 (useExternalStoreId)

Request URL:

<http://esl.zkong.com/zk/erp/item/list?page=1&size=10&useExternalStoreId=1&externalStoreId=100012>

Request Body:

```
{
  "attrClass": "default",
  "attrName": "default"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "list": [
      {
```

```
"id": 11174894570,
"type": 1,
"price": null,
"spec": "",
"productCode": "",
"barCode": "***",
"agencyId": 1629967488587,
"merchantId": 1629968747662,
"storeId": 0,
"originalPrice": null,
"memberPrice": null,
"productArea": "",
"classLevel": "",
"attrCategory": "default",
"attrName": "default",
"sourceCode": "",
"proStartTime": null,
"proEndTime": null,
"unit": null,
"productSku": "",
"createTime": "2023-03-03 14:50:12",
"itemTitle": "***Red Fuji Apple",
"shortTitle": "",
"qrCode": "",
"promotionText": "",
"custFeature1": "",
"custFeature2": "",
...
"custFeature50": "",
"updateTime": "2023-05-30 15:27:14",
"stock1": null,
"stock2": null,
"stock3": null,
"nfcUrl": "",
"storeName": null,
"label": "",
"number": null,
"brandName": null,
"suggestPrice": null,
"coverPic": null,
"delTime": null,
"isSpecs": 0
}
...
],
"totalElements": 150
},
"translate": true,
"originalMessage": null
}
```

3.9 Batch Store Product Import

API Definition

URL	http://[server address]/zk/erp/item/batchStoreImportItem
Example	http://esl.zkong.com/zk/erp/item/batchStoreImportItem
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Import product data into the ESL platform through the HTTP API. 1) When a store imports new products for the first time, the basic product information is synchronized under the merchant. Other stores that have data synchronization enabled can also use this product data. 2) When importing products through the API, if storeId is empty, the products are imported into the overall product list. If storeId is not empty, the product information is imported only into the corresponding store. 3) External store ID is supported for product import. The external store ID must be configured in the ESL system in advance.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
emptyNeedDelete	query	Integer (1)	false	Overwrite rule when product attributes are updated to empty values: 0 keep the original value, 1 overwrite with an empty value (default: 1)
useExternalStoreId	query	Integer (1)	false	Use externalStoreId (0: No 1: Yes) default: 0
merchantId	body	Integer (20)	true	Merchant ID
agencyId	body	Integer (20)	true	Agency ID
unitName	body	Integer (1)	false	Price unit, default: 0 0: Only integers are allowed, length 0-16. Price fields are

					divided by 100 and kept to two decimal places. 1: Supports 14 integer digits and 5 decimal places
timezone		body	String(100)	false	Time zone GMT+08:00/GMT-05:00 (optional; used for automatic price change time conversion)
repricingFullUpdate		body	Integer	false	Whether promotion activities are fully updated. 1: full update (clear previous promotion activities for this product and then add new ones); 0: incremental update (default, appended by approval time - creation time - collection order)
storeItemList		body	List<storeItem> (20000)	true	All-store product list (single request limit: 20,000 products)
	storeId	body	String	false	Store ID/external store ID, controlled by useExternalStoreId. If not provided, import as merchant-level products.
	itemList	body	List<Item> (20000)	true	Product list (single request limit: 20,000 records)
	barCode	body	String (100)	true	Product Code (unique product identifier)
	attrCategory	body	String (100)	true	Template Category

	attrName	body	String (100)	true	Template Attribute
	productCode	body	String (1000)	false	Product Barcode
	productSku	body	String (100)	false	Product SKU
	itemTitle	body	String (500)	false	Product Title
	shortTitle	body	String (500)	false	Short Product Name
	classLevel	body	String (100)	false	Grade
	productArea	body	String (100)	false	Place of Origin
	unit	body	String (100)	false	Sales Unit
	qrCode	body	String (1000)	false	QR Code URL
	nfcUrl	body	String (1000)	false	NFC Link
	spec	body	String (100)	false	Specification
	originalPrice	body	BigDecimal[20,5]	false	Original Price

	price	body	BigDecimal[20,5]	false	Sales Price
	memberPrice	body	BigDecimal[20,5]	false	Member Price
	stock1	body	BigDecimal[10,2]	false	Stock 1
	stock2	body	BigDecimal[10,2]	false	Stock 2
	stock3	body	BigDecimal[10,2]	false	Stock 3
	proStartTime	body	String	false	yyyy-MM-dd HH:mm:ss format
	proEndTime	body	String	false	yyyy-MM-dd HH:mm:ss format
	promotionText	body	String (100)	false	Promotion Text
	custFeature1	body	String (1000)	false	Custom Extended Field 1
	custFeature2	body	String (1000)	false	Custom Extended Field 2
	...	...	...	...	...
	custFeature50	body	String (100)	false	Custom Extended Field 50

		label	body	String (20)	false	Label
		measurementType	body	Integer	false	Measurement type: 1 - by weight, 2 - by count
		listingStatus	body	Integer	false	Listing status: 1 - listed, 2 - delisted
		labelNum	body	String (100)	false	Label Number
		primaryCategory	body	String (100)	false	Primary Product Category
		secondaryCategory	body	String (100)	false	Secondary Product Category
		repricingList	body	List<Item> (30)	false	Automatic Price Change List (maximum: 30)
		repricingType	body	Integer	true	Promotion order type, determines which configuration field group to map to (1 - Price 1, 2 - Price 2, 3 - Price 3)
		auditTime	body	String (50)	false	Approval Time
		repricingDateStart	body	String (50)	true	Promotion start date. Supported formats: yyyy-MM-dd, yyyy/MM/dd, yyyy-MM-dd in Chinese date format, and date ranges separated by ~
		repricingDateEnd	body	String (50)	false	Promotion end date, same format as above

		ateEnd				
		repricingTimeStart	body	String (50)	true	Promotion start time. Supported formats: HH:mm or HH:mm in Chinese time format
		repricingTimeEnd	body	String (50)	true	Promotion end time, same format as above
		price	body	BigDecimal[20,5]		Promotion price, either this field or discount is required
		discount	body	BigDecimal[10,2]		Discount, either this field or price is required
		discountCalculationField	body	String (100)		Discount calculation field required when discount is provided
		copyText	body	String (50)	false	Text

Note: extended fields 1-3 have a length limit of 1000, fields 4-10 have a limit of 500, fields 11-15 have a limit of 200, and fields 16-50 have a limit of 100.

Request Example

Request URL:
<http://esl.zkong.com/zk/erp/item/batchStoreImportItem?useExternalStoreId=1>

Request Body:

```
{
  "agencyId": "1629967488587",
  "merchantId": "1629968747662",
  "unitName": 1,
  "storeItemList": [
    {
      "storeId": "1630484444737",
      "itemList": [
        {
          "attrCategory": "default",
          "attrName": "default",

```

```
"barCode": "100001234",
"itemTitle": "Red Fuji Apple",
"originalPrice": "600",
"price": "6300",
"productArea": "Yantai, Shandong",
"productCode": "200001582",
"productSku": "200001275",
"promotionText": "Promotion Text",
"custFeature1": "Buy five, get one free"
},
{
  "attrCategory": "default",
  "attrName": "default",
  "barCode": "10004562",
  "itemTitle": "Hainan Banana",
  "originalPrice": "10",
  "price": "8",
  "productArea": "Hainan, China",
  "productCode": "200001423",
  "productSku": "20001536",
  "promotionText": "Promotion Text"
}
],
},
{
  "storeId": "1630484444738",
  "itemList": [
    {
      "attrCategory": "default",
      "attrName": "default",
      "barCode": "100001234",
      "itemTitle": "Red Fuji Apple",
      "originalPrice": "600",
      "price": "6300",
      "productArea": "Yantai, Shandong",
      "productCode": "200001582",
      "productSku": "200001275",
      "promotionText": "Promotion Text",
      "custFeature1": "Buy five, get one free"
    },
    {
      "attrCategory": "default",
      "attrName": "default",
      "barCode": "10004562",
      "itemTitle": "Hainan Banana",
      "originalPrice": "10",
      "price": "8",
      "productArea": "Hainan, China",
      "productCode": "200001423",
      "productSku": "20001536",
      "promotionText": "Promotion Text"
    }
  ]
}
```

```
}  
]  
}  
]  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	Product error information see Example 2

Response Example (related error code no.: 3, 10~23)

```
{  
  "success": true,  
  "code": 17002,  
  "data": null,  
  "message": "successful import of merchandise"  
}
```

Example 2

```
{  
  "success": false,  
  "code": 17051,  
  "message": null,  
  "data": {  
    "storeErrorItemVos": [  
      {  
        "storeId": "1630484444738",  
        "errorItemVos": [  
          {  
            "errorMessage": "erp_import_original_price_invalid",  
            "barCode": "100001235"  
          }  
        ]  
      },  
    ],  
  },  
  {  
    "storeId": "1630484444738",
```

```

"errorItemVos": [
{
"errorMessage": "erp_import_price_invalid",
"barCode": "100001234"
}
]
},
"translate": true,
"originalMessage": null
}

```

3.10 Integration Log Query

API Definition

URL	http://[server address]/zk/integratedLog/integratedLogPage
Example	http://esl.zkong.com/zk/integratedLog/integratedLogPage
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query import logs for products, devices, shelves, and other data.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
currentPage	body	Integer	false	Current Page, default: 1
pageSize	body	String	false	pagesize, default: 10
importMode	body	Integer	false	Import method: 1 API import, 2 file import
importType	body	Integer	false	Import type: 1 product, 2 display, 3 ESL 4 AP, 5 shelf, 6 template
status	body	Integer	false	Import result: 1 all successful, 2 partially

				successful 3 all failed
startTime	body	String	false	Start Time (yyyy-MM-ddformat)
endTime	body	String	false	End Time (yyyy-MM-ddformat)
requestId	body	String	false	Request ID
requestIdList	body	List<String>	false	Request ID List

Request Example

Request URL:

<http://esl.zkong.com/zk/integratedLog/integratedLogPage>

Request Body:

```
{
  "status": 1,
  "importMode": 1,
  "importType": 1,
  "requestId": "20250000001",
  "startTime": "2025-06-26",
  "endTime": "2025-07-04",
  "currentPage": 1,
  "pageSize": 10
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Page	

Result Field Definitions:

importModel	Import method: 1 API import, 2 file import
fileName	Import File Name
importType	Import type: 1 product, 2 display, 3 ESL 4 AP, 5 shelf, 6 template
uploadTime	Upload Time

operator	Operator
importTotal	Total Imported
successTotal	Successful Count
errTotal	Failed Count
sourceFilePath	Source File URL
errFilePath	Error File URL
status	Import result: 1 all successful, 2 partially successful 3 all failed
requestId	Request ID

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "totalElements": 1,
    "list": [
      {
        "id": 3,
        "importModel": 1,
        "fileName": null,
        "importType": 1,
        "uploadTime": "2025-07-03 20:23:07",
        "operator": "hxxtest",
        "importTotal": 3,
        "successTotal": 3,
        "errTotal": 0,
        "sourceFilePath": null,
        "errFilePath": null,
        "status": 2,
        "merchantId": 1751544662531,
        "createUser": "66",
        "updateUser": "66",
        "createTime": "2025-07-03 20:23:10",
        "updateTime": "2025-07-03 20:23:10",
        "requestId": "20250001"
      }
    ]
  },
  "translate": true,
  "originalMessage": "success"
}
```

3.11 Batch query product information bound to ESLs (supports shelf dimension).

API Definition

URL	http://[server address]/zk/erp/item/getBindItemList
Example	http://esl.zkong.com/zk/erp/item/getBindItemList
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Batch query product information bound to ESLs (supports shelf dimension).

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Long	false	Store ID (required when external store ID is not used)
barcodes	body	List<String>	true	ESL Barcode List Shelf Number At least one query scope is required. If both are provided, the ESL barcode list takes priority.
shelfNo	body	String		

Request Example

Request URL:

<http://esl.zkong.com/zk/erp/item/getBindItemList>

Request Body:

```
{  
  "storeId": 1694518964214,  
  "barcodes": [  
    "84A778093",
```

```

      "827279656"
    ],
    "shelfNo": "1-2-3"
  }

```

Response Parameters

Parameter Name		Description	Data Type	Notes
success		Request Successful	Boolean	true/false
code		Status Code	Integer	See Appendix 1 for details
message		Message	String	Message
data		Response Data	List<Object>	
	barcode	ESL Barcode	String	
	item	Product Object	Item	

Response Example

```

{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": [
    {
      "barcode": "827279656",
      "item": {
        "id": 65649777,
        "type": 1,
        "price": 11.11,
        "spec": null,
        "productCode": null,
        "barCode": "11",
        "agencyId": 1597317958637,
        "merchantId": 1644307350261,
        "storeId": 1694518964214,
        "originalPrice": 12.12,
        "memberPrice": 13.13,
        "productArea": null,
        "classLevel": null,
        "attrCategory": "default",
        "attrName": "default",

```

```
"sourceCode": null,
"proStartTime": null,
"proEndTime": null,
"unit": null,
"productSku": null,
"createdTime": "2025-04-17 15:03:41",
"itemTitle": "11ceshi",
"shortTitle": null,
"qrCode": null,
"promotionText": null,
"custFeature1": null,
"custFeature2": null,
"custFeature3": null,
"custFeature4": null,
"custFeature5": null,
"custFeature6": null,
"custFeature7": null,
"custFeature8": null,
"custFeature9": null,
"custFeature10": null,
"custFeature11": null,
"custFeature12": null,
"custFeature13": null,
"custFeature14": null,
"custFeature15": null,
"custFeature16": null,
"custFeature17": null,
"custFeature18": null,
"custFeature19": null,
"custFeature20": null,
"custFeature21": null,
"custFeature22": null,
"custFeature23": null,
"custFeature24": null,
"custFeature25": null,
"custFeature26": null,
"custFeature27": null,
"custFeature28": null,
"custFeature29": null,
"custFeature30": null,
"custFeature31": null,
"custFeature32": null,
"custFeature33": null,
"custFeature34": null,
"custFeature35": null,
"custFeature36": null,
"custFeature37": null,
"custFeature38": null,
"custFeature39": null,
"custFeature40": null,
"custFeature41": null,
```

```
"custFeature42": null,
"custFeature43": null,
"custFeature44": null,
"custFeature45": null,
"custFeature46": null,
"custFeature47": null,
"custFeature48": null,
"custFeature49": null,
"custFeature50": null,
"updateTime": "2025-04-22 14:11:01",
"stock1": null,
"stock2": null,
"stock3": null,
"nfcUrl": null,
"storeName": null,
"label": "",
"operationUserId": 73
}
}
},
"translate": true,
"originalMessage": null
}
```

4. Device Operations and Information Query

4.1 Bind Product and ESL

API Definition

URL	http://[server address]/zk/bind/bindItemPriceTag/1
Example	http://esl.zkong.com/zk/bind/bindItemPriceTag/1
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Bind a product and ESL through the HTTP API and perform image refresh.

Request Parameters

Parameter Name	Parameter	Data Type	Required	Notes
----------------	-----------	-----------	----------	-------

Type				
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer(1)	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
itemBarCode	body	String(100)	true	Product Code
priceTagCode	body	String(60)	true	ESL Barcode
shelfNum	body	String(100)	false	Shelf number of the product
apMac	body	String(12)	false	Refresh image using a specified AP (the ESL must be within the AP communication range)

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/bindItemPriceTag/1>
 Request Body:

```
{
  "itemBarCode": "2000123456",
  "priceTagCode": "A0A3B807F374",
  "shelfNum": "",
  "storeId": "1588041156676",
  "apMac": "A0A3B962E350"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for

			details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 37 44 45 47)

```
{
  "success": true,
  "code": 13006,
  "data": null,
  "message": "ESL product binding successful. Image is being issued...",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 15069,
  "message": "Device not found. Please contact the agency.",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.2 Batch Bind ESLs and Products

API Definition

URL	http://[server address]/zk/bind/batchBind
Example	http://esl.zkong.com/zk/bind/batchBind
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Batch bind ESLs and products through the HTTP API.

Request Parameters

Parameter Name	Parameter	Data Type	Required	Notes
----------------	-----------	-----------	----------	-------

Type				
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
tagItemBinds	body	List<Object>	true	Binding Relationship List
eslBarcode	body	String(60)	true	ESL Barcode
itemBarcode	body	String(100)	true	Product Code

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/batchBind>
 Request Body:

```
{
  "storeId": 1588041156676,
  "tagItemBinds": [
    {
      "eslBarcode": "100000886",
      "itemBarcode": "2000123456"
    },
    {
      "eslBarcode": "100000998",
      "itemBarcode": "2000123457"
    }
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message

data	Response Data	Json	
------	---------------	------	--

Response Example (related error code no.: 12 13 20 45)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 11111,
  "message": "Operation failed",
  "data": false,
  "translate": true,
  "originalMessage": "[100000886] esl unavailable"
}
```

4.3 Batch Unbind ESLs

API Definition

URL	http://[server address]/zk/bind/batchUnbind
Example	http://esl.zkong.com/zk/bind/batchUnbind
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Batch unbind ESLs through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
tagItemBinds	body	List<Object>	true	ESL Barcode List
eslBarcode	body	String(60)	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/batchUnbind>
 Request Body:

```
{
  "storeId": 1588041156676,
  "tagItemBinds": [
    {
      "eslBarcode": "100000886"
    },
    {
      "eslBarcode": "100000998"
    }
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 12 13 41 45 48 49 50)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 13030,
  "message": "some esl are unbound",
  "data": [
    "100000886",
    "100000998"
  ],
  "translate": true,
  "originalMessage": null
}
```

4.4 Batch Import Products and Bind ESLs

API Definition

URL	http://[server address]/zk/erp/esl/batchBind
Example	http://esl.zkong.com/zk/erp/esl/batchBind
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Batch bind ESLs through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used

storeId	body	Integer(20)	true	Store ID
unitName	body	Integer(1)	false	Sales Unit default: 0 (0: cent, 1: yuan)
bindList	body	List<Obj>	true	
eslBarcode	body	String(60)	true	ESL Barcode
item	body	Object	true	Product Information
barCode	body	String (100)	true	Product Code (unique product identifier)
attrCategory	body	String (100)	true	Template Category
attrName	body	String (100)	true	Template Attribute
productCode	body	String (100)	false	Product barcode. Separate multiple values with commas
productSKU	body	String (1000)	false	Product SKU
itemTitle	body	String (500)	false	Product Title
shortTitle	body	String (500)	false	Short Product Name
classLevel	body	String (100)	false	Grade
productArea	body	String (100)	false	Place of Origin

	unit	body	String (100)	false	Sales Unit
	qrCode	body	String (1000)	false	QR Code URL
	nfcUrl	body	String (1000)	false	NFC Link
	spec	body	String (100)	false	Specification
	originalP rice	body	BigDecimal[20,5]	false	Original Price
	price	body	BigDecimal[20,5]	false	Sales Price
	member Price	body	BigDecimal[20,5]	false	Member Price
	stock1	body	BigDecimal[10,2]	false	Stock 1
	stock2	body	BigDecimal[10,2]	false	Stock 2
	stock3	body	BigDecimal[10,2]	false	Stock 3
	proStart Time	body	String	false	yyyy-MM-dd HH:mm:ss format
	proEndTi	body	String	false	yyyy-MM-dd HH:mm:ss

me					format
promoti onText	body	String (100)	false		Promotion Text
custFeat ure1	body	String (1000)	false		Custom Extended Field 1
custFeat ure2	body	String (1000)	false		Custom Extended Field 2
...	...	...	...		...
custFeat ure50	body	String (100)	false		Custom Extended Field 50
label	body	String (20)	false		Label

Note: extended fields 1-2 have a length limit of 1000, fields 3-5 have a limit of 500, and fields 6-50 have a limit of 100.

Request Example

Request URL:
<http://esl.zkong.com/zk/erp/esl/batchBind>
 Request Body:

```
{
  "storeId": "1591878736794",
  "unitName": 1,
  "bindList": [
    {
      "eslBarcode": "100000886",
      "item": {
        "attrCategory": "default",
        "attrName": "default",
        "barCode": "100001234",
        "itemTitle": "Red Fuji Apple",
        "originalPrice": "600",
        "price": "6300",
        "productArea": "Yantai, Shandong",
        "productCode": "200001582",
        "productSku": "200001275",
        "promotionText": "Promotion Text"
      }
    }
  ]
}
```

```
}  
},  
{  
  "eslBarcode": "100000887",  
  "item": {  
    "attrCategory": "default",  
    "attrName": "default",  
    "barCode": "100001254",  
    "itemTitle": "Hainan Banana",  
    "originalPrice": "10",  
    "price": "8",  
    "productArea": "Hainan, China",  
    "productCode": "200001423",  
    "productSku": "20001536",  
    "promotionText": "Promotion Text"  
  }  
}  
]  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 1 3 11 12 13 45 49 51)

```
{  
  "success": true,  
  "code": 100000,  
  "data": null,  
  "message": "Operation successful",  
  "translate": true,  
  "originalMessage": ""  
}
```

Example 2

```
{
  "success": false,
  "code": 11111,
  "message": "Operation failed",
  "data": false,
  "translate": true,
  "originalMessage": "[100000886] esl unavailable"
}
```

4.5 Multi-product Template Binding

API Definition

URL	http://[server address]/zk/bind/bindMulti
Example	http://esl.zkong.com/zk/bind/bindMulti
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Multi-product Template Binding

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
BindMultiRo	body	Object	true	Binding Parameters
storeId	body	Integer(20)	true	Store ID
priceTagCode	body	String(60)	true	ESL Barcode
shelfNum	body	String(60)	false	Shelf Number
itemOrders	body	List<Object> (10)	true	Product Code List
itemCode	body	itemCode(100)	true	Product Code
order	body	Integer(2)	true	Product barcode

Request Example

```
Request URL:
http://esl.zkong.com/zk/bind/bindMulti
Request Body:
{
  "itemOrders": [
    {
      "itemCode": "12001445",
      "order": 1
    }
  ],
  "priceTagCode": "100005939",
  "storeId": 1584608946868,
  "templateId": 1523
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 12 13 27 41 46 47 52 53 54 55 56 57)

```
{
  "success": true,
  "code": 13006,
  "data": null,
  "message": "ESL product binding successful. Image is being issued...",
  "translate": true,
  "originalMessage": "The template size or color does not match the ESL."
}
```

Example 2

```
{
  "success": false,
  "code": 13027,
  "message": "The template size does not match the ESL.",
  "data": null,
  "translate": true,
  "originalMessage": "The template size or color does not match the ESL."
}
```

4.6 Invert ESL Display Image

API Definition

URL	http://[server address]/zk/bind/reverseShow
Example	http://esl.zkong.com/zk/bind/reverseShow
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Invert the ESL display image by ESL barcode.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
barcode	query	String(60)	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/reverseShow?barcode=10060001>
 Request Body:
 None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 44 55 58 59 60)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": ""
}
```

4.7 Force Refresh by Barcode

API Definition

URL	http://[server address]/zk/bind/updateForceByBarCodes
Example	http://esl.zkong.com/zk/bind/updateForceByBarCodes
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Force refresh ESLs through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
barCodeList	body	List<String>	false	ESL barcode list (if empty, all ESLs under the store are force refreshed)
storeId	Integer	Integer(20)	true	Store ID

Request Example

Request URL:
`http://esl.zkong.com/zk/bind/updateForceByBarCodes?useExternalStoreId=1&externalStoreId=123456`

Request Body:

```
{
  "storeId": 1605147967965,
  "barCodeList": [
    "100854525",
    "100854565"
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 42 45 61)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 13040,
  "message": "Binding relationship does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.8 Execute Flashing by Barcode

API Definition

URL	http://[server address]/zk/led/executeLedByBarCodes
Example	http://esl.zkong.com/zk/led/executeLedByBarCodes
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Execute ESL flashing through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
storeId	body	Integer (20)	false	Store ID
externalStoreId	body	String (40)	false	External Store ID
useExternalStoreId	body	Integer	false	Use externalStoreId 1 - use, 0 - do not use; if not provided, do not use by default
lightColor	body	Integer (1)	false	flashcolor (default:red) 1-8 (redgreenblueyelloworange cyanpurplewhite)
lightTime	body	Integer (8)	true	flashduration Unit: seconds; maximum duration: 4 hours

brightness	body	Integer (3)	false	flashbrightness(0-100), default: 40%
lightFrequency	body	Integer (1)	false	flashfrequency1: slow flash (default) 2: fast flash
barCodes	body	List<String>	false	ESL Barcode List
itemBarCodes	body	List<String>	false	Product Code List (when both product code and ESL barcode are provided, only the ESL barcode takes effect)
serialNum	body	Integer	false	LED index for turning off/flashing, starting from 0 (picking ESL)
flashType	body	Integer	false	Flashing type: 0 normal flashing, 1 steady on (picking ESL)

Request Example

Request URL:

<http://esl.zkong.com/zk/led/executeLedByBarCodes>

Request Body:

```
{
  "storeId": 1605147967965,
  "externalStoreId": "wb001",
  "itemBarCodes": [
    "123456"
  ],
  "barCodes": [
    "A0A3B82001DD"
  ],
  "lightTime": "60",
  "lightFrequency": "1",
  "lightColor": "1"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 19 38 62 63)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

4.9 Scheduled ESL Flashing API

API Definition

URL	http://[server address]/zk/led/timeFlash
Example	http://esl.zkong.com/zk/led/timeFlash
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Schedule flashing for ESLs or products under a store using cron expressions.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	false	Store ID
itemBarcodes	body	List<String>	false	Product Barcode List

pricetagBarcode	body	List<String>	false	ESL Barcode List (when both product code and ESL barcode are provided, only the ESL barcode takes effect)
startFlashCron	body	String	false	startflashcron expression
stopFlashCron	body	String	false	Start flashing cron expression
lightFrequency	body	Integer(1)	false	Flashing frequency 1: slow flash 2: fast flash default: slow flash
lightColor	body	Integer(1)	true	flashcolor1-8 (redgreenblueyelloworangecyan purplewhite)
brightness	body	Integer(3)	true	flashbrightness(0-100) default: 40

Request Example

Request URL:
<http://esl.zkong.com/zk/led/timeFlash>

Request Body:

```
{
  "storeId": null,
  "useExternalStoreId": 1,
  "externalStoreId": "00001",
  "itemBarcodes": [
    "ABCD",
    "ABC"
  ],
  "startFlashCron": "* * 9 * * ?",
  "stopFlashCron": "* * 30 9 * * ?",
  "lightFrequency": 1,
  "lightColor": 1,
  "brightness": 40
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 12 13 62 64 65 66 67)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.10 Stop Flashing by Product Code

API Definition

URL	http://[server address]/zk/led/stopLedByItem
Example	http://esl.zkong.com/zk/led/stopLedByItem
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Stop flashing for ESLs bound to the product code.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
serialNum	query	Integer	false	LED index for turning off/flashing, starting from 0 (picking ESL)

flashType	query	Integer	false	Flashing type: 0 normal flashing, 1 steady on (picking ESL)
storeId	query	Integer(20)	true	Store ID
barcodes	body	List<String>	true	Product Code List

Request Example

Request URL:
<http://esl.zkong.com/zk/led/stopLedByItem?storeId=1588041156676>
 Request Body:

```
[
  "2000123456",
  "2000135798"
]
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	

Response Example (related error code no.: 12 13 38 45 63)

```
{
  "success":true,
  "code": 10000,
  "data": "",
  "message":null,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 40005,
  "message": "ESL does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.11 Stop Flashing by ESL Barcode

API Definition

URL	http://[server address]/zk/led/stopLedByPricetag
Example	http://esl.zkong.com/zk/led/stopLedByPricetag
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Stop flashing for the specified ESL by ESL barcode.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	query	Integer(20)	true	Store ID
serialNum	query	Integer	false	LED index for turning off/flashing, starting from 0 (picking ESL)
flashType	query	Integer	false	Flashing type: 0 normal flashing, 1 steady on (picking ESL)

barcodes	body	List<String>	true	ESL Barcode List
----------	------	--------------	------	------------------

Request Example

Request URL:

<http://esl.zkong.com/zk/led/stopLedByItem?storeId=1588041156676>

Request Body:

```
[
  "2000123456",
  "2000135798"
]
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	

Response Example (related error code no.: 12 13 38 45 63)

```
{
  "success":true,
  "code": 10000,
  "data": "",
  "message":null,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success":false,
  "code": 11111,
  "data": "A0A3B8274380: The ESL does not belong to this merchant",
  "message": null,
  "translate": true,
}
```

```
"originalMessage": null
}
```

4.12 Batch Reboot ESLs

API Definition

URL	http://[server address]/zk/pricetag/eslReboot
Example	http://esl.zkong.com/zk/pricetag/eslReboot
Request Method	POST
Content Type	application/json;charset=utf-8
Description	When storeId is provided, all ESLs under the current store will be rebooted. When storeId is not provided and barcodeList is provided, only the specified ESLs in barcodeList will be rebooted.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	query	Integer(20)	false	Store ID
barcodeList	body	List<String>	false	ESL Barcode List

Request Example (reboot three ESLs)

Request URL:
<http://esl.zkong.com/zk/pricetag/eslReboot>

Request Body:

```
{
  "barcodeList": [
    "A0A3B8273D71",
    "A0A3B8273D7C",
    "A0A3B8273D7C"
  ]
}
```

```
}  
}
```

(reboot all ESLs under the specified store)

Request URL:

<http://esl.zkong.com/zk/pricetag/eslReboot>

Request Body:

```
{  
  "storeId": 123456789  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13)

```
{  
  "success": true,  
  "code": 15011,  
  "message": "Reboot message has been sent",  
  "data": null,  
  "translate": true,  
  "originalMessage": null  
}
```

4.13 Add AP to Store

API Definition

URL	http://[server address]/zk/erp/ap/add
Example	http://esl.zkong.com/zk/erp/ap/add
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Add an AP device to the specified store.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
mac	60	String(60)	true	AP MAC Address
apName	60	String(60)	false	AP Name
comment	100	String(100)	false	Remarks

Request Example

Request URL:
<http://es1.zkong.com/zk/erp/ap/add>
 Request Body:

```
{
  "storeId": "1588041156676",
  "mac": "A0A3B80C2BE0",
  "apName": "Tennstedt-AP",
  "comment": ""
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 1 68 69)

```
{
  "success":true,
  "code":15001,
  "data": "",
  "message": "AP added successfully",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 15060,
  "message": "Device not found",
  "data": null,
  "translate": true,
  "originalMessage": "AP device not found"
}
```

4.14 Delete AP device.

API Definition

URL	http://[server address]/zk/erp/ap/del
Example	http://esl.zkong.com/zk/erp/ap/del
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Delete AP device.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID

mac	body	String(60)	true	APMAC
-----	------	------------	------	-------

Request Example

Request URL:

<http://es1.zkong.com/zk/erp/ap/del>

Request Body:

```
{
  "storeId": "1588041156676",
  "mac": "A0A3B8273D77"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 49)

```
{
  "success":true,
  "code":15002,
  "data": "",
  "message": "AP deleted successfully",
  "translate": true,
  "originalMessage": "AP device not found"
}
```

Example 2

```
{
  "success": false,
  "code": 15060,
  "message": "Device not found",
  "data": null,
  "translate": true,
  "originalMessage": "AP device not found"
}
```

4.15 Query AP status information.

API Definition

URL	http://[server address]/zk/erp/ap/list
Example	http://esl.zkong.com/zk/erp/ap/list
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query AP status information.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
page	query	Integer	false	Default: 0(page 1)
size	query	Integer	false	Default: 10
storeId	query	Integer(20)	false	Store ID (if not provided, query all data under the merchant)
mac	query	String(60)	false	APMAC

Request Example

Request URL:

<http://esl.zkong.com/zk/erp/ap/list?storeId=1594107807481>

Request Body:

None

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	
Response Data	data	Json	
Total Records	totalElements	Integer	
AP Information List	list	List	
Store ID	storeId	Integer	
Merchant ID	merchantId	Integer	
Agency ID	agencyId	Integer	
APID	id	String	AP ID in the ESL system
Device MAC	mac	String	
Manufacturer	manufacturer	String	
Device Model	model	String	
AP Name	apName	String	
SN	sn	String	

Online Status	online	Integer	0: offline, 1: online, 2: upgrading, 3: upgrade failed
Notes	comment	String	
Operation Status	operstate	Integer	
Software Version Number	softVersion	String	
Hardware Version Number	hardVersion	String	
Added Time	addTime	String	
Deleted Time	delTime	String	
Online Time	joinTime	String	
Offline Time	offLineTime	String	
Role Code	onlineTime	Integer	
Number of Associated ESLs	eslCount	Integer	
Whether the Data Is Valid	flag	String	0: invalid, 1: valid
Upgrade Time	upgradeTime	String	
2G Wi-Fi Name	wifiName2G	Integer	
5G Wi-Fi Name	wifiName5G	String	
Wi-Fi Setting Status	wifiStatus	String	

	Wi-Fi Setting Time	wifiSetTime	String	
	Upgrade Status	upgradeState	Integer	
	Base station IP address	ip	String	
	Work Mode	workMode	Integer	Image refresh mode is 1; normal mode is 0
	Reboot Status	rebootState	Integer	0: successful, 1: issuing, 2: failed
	Reboot Time	rebootTime	String	Last Reboot Time
	First Online Time	firstJoinTime	String	First time in current store

Response Example (related error code no.: 12 13 45)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "list": [
      {
        "id": 1966,
        "mac": "A0A3B8167870",
        "manufacturer": null,
        "model": "ZAP-DL",
        "apName": "A0:A3:B8:16:78:70",
        "sn": null,
        "online": 0,
        "comment": "",
        "operstate": null,
        "agencyId": null,
        "merchantId": 1676959167307,
        "storeId": 1736569498508,
        "softVersion": "6.2.007_Beta20241217",
        "hardVersion": "5",
        "addTime": "2025-02-13 16:06:18",
        "delTime": null,
      }
    ]
  }
}
```

```
"firstJoinTime": "2025-02-26 15:35:19",
"joinTime": "2025-02-13 16:06:18",
"offLineTime": null,
"onlineTime": "0",
"eslCount": 0,
"flag": null,
"upgradeTime": null,
"wifiName2G": null,
"wifiName5G": null,
"wifiStatus": null,
"wifiSetTime": null,
"upgradeState": null,
"storeName": null,
"merchantName": null,
"workMode": 0,
"ip": "192.168.17.75",
"rebootState": 0,
"rebootTime": "2025-02-13 15:27:07",
"elementAttribute": null,
"yawAngle": 295.9,
"pitchAngle": -165.9,
"rollAngle": 174.8,
"supportLocation": null,
"updateTime": "2025-02-26 15:35:21",
"lcdCount": null
}
],
"totalElements": 1
},
"translate": true,
"originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 14031,
  "data": null,
  "message": "Insufficient permissions"
}
```

4.16 Query ESL status information.

API Definition

URL	http://[server address]/zk/erp/esl/list
Example	http://esl.zkong.com/zk/erp/esl/list
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query ESL status information.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
page	query	Integer	false	Page Number Default: 0(page 1)
size	query	Integer	false	Page Size Default: 10
searchBindRo	body	Object		Query condition object
storeId	body	Integer(20)	false	Store ID (if not provided, query all data under the merchant)
itemBarCode	body	String(60)	false	Product Code
itemTitle	body	String(500)	false	Product Name
priceTagCode	body	String(60)	false	ESL Barcode
oemModel	body	String(20)	false	ESL Model
shelfNo	body	String(60)	false	Shelf Number

Request Example

Request URL:

<http://esl.zkong.com/zk/erp/esl/list?page=1&size=10>

Request Body:

```
{
  "storeId": 1571661761656,
  "itemBarCode": "",
  "itemTitle": "",
  "priceTagCode": "",
  "oemModel": "",
  "shelfNo": ""
}
```

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	
Response Data	data	Json	
Total Records	totalElements	Integer	
ESL Information List	list	List	
Store ID	storeId	Integer	
Merchant ID	merchantId	Integer	
Agency ID	agencyId	Integer	
ESL ID	id	Integer	ESL ID in the ESL system
ESL Barcode	priceTagCode	String	

Product Code	itemBarCode	String	Product code bound to the ESL
Width	width	Integer	ESL width resolution
Height	height	Integer	ESL height resolution
Remaining Battery	battery	Integer	1-100
Battery Level	batteryLevel	Integer	1-5
ESL Model	oemModel	String	
status	state	Integer	0: offline, 1: online
Binding Status	bindState	Integer	0: unbound, 1: bound
Binding Time	bindTime	String	
Current Signal Strength	apSignal	Integer	Current AP Signal Strength
End time	updateTime	String	Binding Update Time
Last Communication Time	lastCommuTime	String	
Software Version	softVersion	String	
Shelf Number	shelfNum	String	

Response Example (related error code no.: 12 13 45)

```
{
```

```
"success": true,
"code": 13001,
"message": "Binding list queried successfully",
"data": {
  "list": [
    {
      "id": 2,
      "agencyId": 1591927217946,
      "merchantId": 1595819724137,
      "storeId": 1595819724189,
      "priceTagCode": "18BC5A6DD32C",
      "itemBarCode": "100001254",
      "bindTime": "2020-07-27 15:42:09",
      "updateTime": "2020-07-27 16:42:09",
      "itemTitle": "iPhone 11 Pro",
      "oemModel": "ZKC21S",
      "softVersion": "2.1.18",
      "state": 1,
      "batteryLevel": 5,
      "lastCommuTime": "2020-07-28 14:09:59",
      "shelfNo": null,
      "width": 250,
      "height": 128,
      "battery": 100,
      "bindState": 1,
      "apSignal": 42
    },
    {
      "id": 1,
      "agencyId": 1591927217946,
      "merchantId": 1595819724137,
      "storeId": 1595819724189,
      "priceTagCode": "18BC5A6DD0D6",
      "itemBarCode": "100001234",
      "bindTime": "2020-07-27 14:20:04",
      "updateTime": "2020-07-27 15:20:04",
      "itemTitle": "HuaWei P40",
      "oemModel": "ZKC21S",
      "softVersion": "2.1.19",
      "state": 1,
      "batteryLevel": 5,
      "lastCommuTime": "2020-07-27 15:20:32",
      "shelfNo": null,
      "width": 250,
      "height": 128,
      "battery": 100,
      "bindState": 1,
      "apSignal": 50
    }
  ],
  "totalElements": 2,
```

```

    "translate": true,
    "originalMessage": null
  }
}

```

4.17 Query ESL Binding Information by Barcode (with preview image)

API Definition

URL	http://[server address]/zk/bind/viewBindPicByBarcode
Example	http://esl.zkong.com/zk/bind/viewBindPicByBarcode
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query ESL binding information by barcode. The returned image preview is in Base64 format and the variable name is images.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
barcode	query	String(60)	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/viewBindPicByBarcode?barcode=8002087682>
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message

data	Response Data	Json	
------	---------------	------	--

Response Example (related error code no.: 41)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "updateTime": "2022-08-02 16:03:43",
    "type": 1,
    "tagBarCode": "800208768",
    "softVersion": "2.2.30",
    "state": 1,
    "model": "LT26V-N",
    "size": "2.6",
    "battery": 1,
    "batteryLevel": 1,
    "lastCommuTime": "2023-05-25 07:00:12",
    "shelfNum": null,
    "deviceType": 1,
    "apSignal": null,
    "turnOver": 0,
    "itemBindInfos": [
      {
        "id": null,
        "itemOrder": 1,
        "itemBarCode": "1",
        "itemTitle": "cola330ml",
        "templateAttr": "default"
      }
    ],
    "images": [
      "Qk12DwIAAAAAADYAAAAoAAAAKAEEAJgAAAABABgAAAAAAEAP
      AgAAAAAAAAAAAAAAAAAAAAA////////\\r\\n//////////....."
    ],
    "segmentCodeViewVo": null
  },
  "translate": true,
  "originalMessage": null
}
```

Example 2 (usually the provided barcode is not bound or does not exist)

```
{
  "success": false,
  "code": 10009,
  "message": "Parameter error",
  "data": null,
  "translate": true,
  "originalMessage": "param invalid"
}
```

4.18 Query All Bound ESL Details by Product Code

API Definition

URL	http://[server address]/zk/bind/getPriceTagsByItemBarCode
Example	http://esl.zkong.com/zk/bind/getPriceTagsByItemBarCode
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Get all bound ESL information by product code.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
itemBarCode	body	String(100)	true	Product Code
externalStoreId	body	String(40)	true	External Store ID

Request Example 1

Request URL:
http://esl.zkong.com/zk/bind/getPriceTagsByItemBarCode

Request Body:

```
{  
  "itemBarCode": "123456",  
  "externalStoreId": "123456"  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 27 42)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "item": {
      "itemBarCode": "123456789",
      "itemTitle": "Red Fuji Apple",
      "itemPrice": 3
    },
    "priceTags": [
      {
        "id": 391635,
        "barCode": "A0A3B8204C96",
        "mac": "A0A3B8204C96",
        "apMac": "A0A3B80D23D0",
        "apList": "{\"A0A3B88F5A00\":61,\"A0A3B80ED200\":53,\"A0A3B80AAFA0\":34,\"A0A3B88F5990\":27,\"A0A3B80B9DB0\":27,\"A0A3B8700A00\":37}",
        "apSignal": null,
        "shelfNum": null,
        "width": 250,
        "height": 128,
        "model": "ZKC21S",
        "state": 0,
        "bindState": 1,
        "upgradeState": 5,
        "lastCommuTime": "2021-09-23 17:11:20",
        "battery": 100,
        "batteryLevel": 5,
        "deviceType": 1,
        "agencyId": 2147483648,
        "merchantId": 1615365123378,
        "storeId": 1615365123392,
        "sequence": 0,
        "accessKey": null,
        "currentTime": 7586,
        "currentSysTime": 1632388280,
        "upgradeTime": "2021-05-27 11:13:36",
        "lastSendStartTime": null,
        "lastReceiveAckTime": null,
        "addTime": "2021-05-17 14:35:01",
        "sendState": null,
        "softVersion": "2.1.34",
        "hardVersion": null,
        "picNum": 13,
        "turnOver": 0,
        "rebootStatus": 2,

```

```
"chipSolution": 0,  
"abilities": null  
},  
  "translate": true,  
  "originalMessage": null  
}
```

Example 2

```
{  
  "success": false,  
  "code": 19015,  
  "message": "No store matches the external store ID",  
  "data": null,  
  "translate": true,  
  "originalMessage": null  
}
```

4.19 Get bound ESL information under a store.

API Definition

URL	http://[server address]/zk/bind/findBinds/{page}/{size}
Example	http://esl.zkong.com/zk/bind/findBinds/1/0
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Get bound ESL information under a store.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
page	path	Integer	true	Page Number

size	path	Integer	true	Page Size
storeIds	body	List<Long>	false	Store ID list; if not provided, all stores of the current user are used
model	body	String(20)	false	ESL Model (if not provided, query all)
state	body	Integer(1)	false	ESL status (0: offline, 1: online)
deviceType	body	Integer(1)	false	Color (0: black/white, 1: black/white/red, 2: black/white/yellow, 3: multi-color)
batteryLevel	body	Integer(1)	false	ESL battery level (1: urgently needs charging, 2: low battery, 3: medium battery, 4: high battery, 5: full battery)
turnOver	body	Integer(1)	false	Whether inverted (0: no, 1: yes)
apSignalStart	body	Integer	false	Minimum Signal Value
apSignalEnd	body	Integer	false	Maximum Signal Value
priceTagCode	body	String(60)	false	ESL Barcode
itemBarCode	body	String(100)	false	Product Code
itemTitle	body	String(100)	false	Product Name
shelfNo	body	String(60)	false	Shelf Number
labelList	body	List<String>	false	Label List
orgIdList	body	List<Integer>	false	Organization ID List

Request Example 1

Request URL:
<http://esl.zkong.com/zk/bind/findBinds/1/0>
Request Body:

```
{
  "model": "ZKC13F-E4",
  "state": "1",
  "deviceType": "1",
  "batteryLevel": "5",
```

```
"turnOver": "0",  
"storeIds": [  
  
],  
"priceTagCode": null,  
"itemBarCode": null,  
"itemTitle": null,  
"shelfNo": null,  
"apSignalStart": 2,  
"apSignalEnd": 88,  
"labelList": null,  
"orgIdList": null  
}
```

Request Example 2: use externalStoreId

Request URL:

<http://es1.zkong.com/zk/bind/findBinds/1/0?useExternalStoreId=1&externalStoreId=12345>

Request Body:

```
{  
  "model": "ZKC13F-E4",  
  "state": "1",  
  "deviceType": "1",  
  "batteryLevel": "5",  
  "turnOver": "0",  
  "storeIds": [],  
  "priceTagCode": null,  
  "itemBarCode": null,  
  "itemTitle": null,  
  "shelfNo": null,  
  "apSignalStart": 2,  
  "apSignalEnd": 88  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Data Field Definitions

bindType	bindtype 1 single-product 2multi-product
templateId	bindtemplateID
bindTime	Binding Time
priceTagCode	ESL Barcode
model	ESL Model
softVersion	Firmware Version
state	Status: 0 offline, 1 online
battery	Battery
lastCommuTime	Last Communication Time
apSignal	Apsignal
deviceType	Device Type
turnOver	Whether inverted: 0 no, 1 yes
apList	nearbyaplist
apMac	Last Used AP
storeId	Store ID
externalStoreId	Store code
storeName	Store Name
organization	Organization
itemBindInfos	Bound Product Information
itemBarCode	Product Code
itemTitle	Product Name
templateAttr	Template Attribute
templateCategory	Template Category
label	Product Label
originalPrice	Original Price
price	Sales Price

Response Example (related error code no.: 12 13 45)

```
{
  "success": true,
  "code": 13001,
  "message": "Binding list queried successfully",
  "data": {
    "list": [
      {
        "id": 343231,
        "bindType": 1,
        "templateId": 10344,
        "bindTime": "2021-09-09 16:45:56",
        "updateTime": "2021-09-13 14:58:50",
        "priceTagCode": "A0A3B8204C96",
        "model": "ZKC21S",
        "softVersion": "2.1.34",
        "state": 0,
        "battery": 100,
        "batteryLevel": 5,
        "lastCommuTime": "2021-09-23 17:11:20",
        "apSignal": 60,
        "shelfNo": null,
        "deviceType": 1,
        "turnOver": 0,
        "noData": 0,
        "storeId": 17251461154,
        "storeName": "Hangzhou Qiantang District Baiyang Street Store",
        "externalStoreId": 10005,
        "organization": "Hangzhou",
        "apMac": "A0A3B80D23D0",
        "apList": "{\\\"A0A3B88F5A00\\\":61,\\\"A0A3B80ED200\\\":53,\\\"A0A3B80AAFA0\\\":34,\\\"A0A3B88F5990\\\":27,\\\"A0A3B80B9DB0\\\":27,\\\"A0A3B8700A00\\\":37}",
        "itemBindInfos": [
          {
            "id": 1000418055,
            "itemOrder": 0,
            "itemBarCode": "123456789",
            "itemTitle": "Yantai Red Fuji Apple",
            "originalPrice": "6.75",
            "price": "6.25",
            "label": "Apple",
            "templateAttr": "promotion",
            "templateCategory": "Normal"
          }
        ]
      }
    ]
  },
}
```

```
"totalElements": 1,  
  "translate": true,  
  "originalMessage": null  
}  
}
```

Example 2

```
{  
  "success": false,  
  "code": 19017,  
  "message": "At least one of external store ID and store ID must not be empty",  
  "data": null,  
  "translate": true,  
  "originalMessage": null  
}
```

4.20 Query information for a single bound ESL under a merchant.

API Definition

URL	http://[server address]/zk/bind/geyPriceTagByPriceTagBarCode
Example	http://esl.zkong.com/zk/bind/geyPriceTagByPriceTagBarCode
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query information for a single bound ESL under a merchant.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
priceTagBarCode	body	String	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/geyPriceTagByPriceTagBarCode>

Request Body:
{

```
"priceTagBarCode": "A0A3B8204C96"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: None)

```
{
  "success": true,
  "code": 15067,
  "message": "ESL information queried successfully",
  "data": {
    "tagBarCode": "A0A3B8204C96",
    "shelfNum": null,
    "itemBarCode": null,
    "itemTitle": null,
    "itemPrice": null,
    "state": 0,
    "upgradeState": 5,
    "softVersion": "2.1.34",
    "model": "ZKC21S",
    "lastCommuTime": "2021-09-23 17:11:20",
    "id": 391635,
    "mac": "A0A3B80D23D0",
    "apSignal": null,
    "battery": 100,
    "batteryLevel": 5,
    "storeName": "hsb",
    "tagMac": "A0A3B8204C96",
    "items": [
      {
        "itemBarCode": "123456789",
        "itemTitle": "Red Fuji Apple",
        "itemPrice": 3
      }
    ]
  }
}
```

```

    ],
    "translate": true,
    "originalMessage": null
  }

```

Example 2

```

{
  "success": false,
  "code": 13040,
  "message": "Binding relationship does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}

```

4.21 Query Image Refresh Logs

API Definition

URL	http://[server address]/zk/erp/log/listLog
Example	http://esl.zkong.com/zk/erp/log/listLog
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query store image refresh logs.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
page	query	Integer	true	Page number, default: 0 (first page)

size	query	Integer	true	Items per page (default: 10, maximum: 2000)
storeId	body	Long(20)	false	Store ID
createdTime	body	String	true	Log creation time range, e.g. 2022-4-21,2022-4-27 The first value is the start time; the second is the end time
operation	body	Integer(2)	false	Operation Type
status	body	Integer(1)	false	Image Refresh Result
priceTagBarCode	body	String(60)	false	ESL Barcode
itemBarCode	body	String(100)	false	Product Code
itemName	body	String(100)	false	Product Name
shelfNo	body	String(60)	false	Shelf Number
feedBackTimeOrder	body	String(4)	false	End time sort asc - ascending desc - descending Select one of the two sorting criteria
timeCostOrder	body	String(4)	false	Duration sort asc - ascending desc - descending Select one of the two sorting criteria

Request Example

Request URL:

<http://esl.zkong.com/zk/erp/log/listLog?page=0&size=10>

Request Body:

```
{
  "storeId": 1632276202577,
  "createdTime": "2022-4-21,2022-4-27",
  "operation": null,
  "model": null,
}
```

```
"status": null,
"priceTagBarCode": null,
"itemBarCode": null,
"itemName": null,
"shelfNo": null,
"feedBackTimeOrder": "asc",
"timeCostOrder": null
}
```

Example 2

```
http://esl.zkong.com/zk/erp/log/listLog?page=1&size=10&useExternalStoreId=1&externalStoreId=default
{
"createdTime": "2022-4-21,2022-4-27",
"operation":null,
"model": null,
"status": null,
"priceTagBarCode": null,
"itemBarCode": null,
"itemName": null,
"shelfNo": null,
"feedBackTimeOrder": "asc",
"timeCostOrder": null
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 3 12 13 41)

```
{
"success": true,
"code": 10000,
"message": "Operation successful",
"data": {
"list": [
{
"id": 16902,
"productId": null,

```

```

"itemBarCode": "00000",
"itemName": null,
"price": "1569.00000",
"itemAttrName": null,
"priceTagBarCode": "A0A3B8717273",
"deviceType": 5,
"model": "ZKE73B-N",
"shelfNo": null,
"operation": 9,
"status": 3,
"operator": "jsk123",
"timeCost": null,
"pushTime": "2022-04-21 05:00:03",
"feedbackTime": "2022-04-21 05:00:03",
"taskId": null,
"returnCode": null,
"agencyId": null,
"merchantId": null,
"storeId": null,
"createdTime": "2022-04-21 05:00:03",
"itemOrder": 1,
"itemLogInfos": [
{
  "id": null,
  "itemOrder": null,
  "itemBarCode": "00000",
  "itemTitle": null,
  "templateAttr": null,
  "price": 1569
}
]
...
],
"totalElements": 22
},
"translate": false
}

```

Note:

Operation Type (operation)

1	Bind
2	Unbind
3	Force Image Refresh
4	Product Change
5	Template Change
13	Triggered Image Refresh
14	Smart Resend

Image Refresh Result (status)

2	Success
3	Failed - Timeout
4	Failed - Timeout Retry
5	Failed - Terminated
7	Manual Retry
8	Failed - No Template
9	Failed - ESL Offline
10	Failed - No Available AP
12	Failed - Message Processing Failed
13	Failed - ESL Not Reported
14	Success - Image Information Unchanged
15	Failed - No Image Information
16	ESL Data Error
17	Send Data Error

4.22 Query ESL Information

API Definition

URL	http://[server address]/zk/pricetag/listPriceTagRelevantInfo
Example	http://esl.zkong.com/zk/pricetag/listPriceTagRelevantInfo
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query ESL-related information.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
currentPage	body	Integer	true	Current Page (starting from 1)
pageSize	body	Integer	true	Page Size
relationallInfo	body	Object	false	Query store-related information.

					When both values are passed, externalStoreId takes effect. When neither is passed, ESL information for the merchant of the current user is queried.
	storeId	body	Long	false	Store ID
	externalStoreId	body	String	false	External Store ID
	priceTagInfo	body	Object	false	Query ESL-related information.
	barCodeList	body	List<Long>	false	storeIdlist
	oemModelList	body	List<String> >	false	externalStoreId list
	state	body	Integer	false	Status (0: offline, 1: online)
	deviceType	body	Integer	false	Device type (0: black/white, 1: black/white/red, 2: black/white/yellow)
	shelfNoList	body	List<String> >	false	Shelf Number List
	apSignalStart	body	Integer(2)	false	AP signal start range
	apSignalEnd	body	Integer(2)	false	AP signal end range
	batteryLevel	body	Integer(1)	false	Battery level (3: normal, 2: low battery, 1: urgent)
	turnOver	body	Integer(1)	false	Flip ESL display image (0: normal, 1: flipped)
	bindState	body	Integer(1)	false	Binding Status(0:Unbind 1:Bind)
	itemQueryCondition	body	Object	false	Query product-related information When both barCode and itemTitle are passed, only barCode takes effect.
	barCode	body	String(100)	false	Product Code
	itemTitle	body	String(100)	false	Product Name

displayInfo	body	Object	true	Information to display in the returned result. It is not allowed to exclude ESL information. It is not allowed to query product information only without querying binding relationship information.
needPriceTagInfo	body	Boolean	true	Whether to display ESL information. Default: true.
needBindInfo	body	Boolean	false	Whether to display binding relationship information
needItemInfo	body	Boolean	false	Whether to display product information associated with the ESL
needRelationalInfo	body	Boolean	false	Whether to display store information

Request Example

Request URL:

<http://esl.zkong.com/zk/pricetag/listPriceTagRelevantInfo>

Request Body:

```
{
  "relationalInfo": {
    "storeId": 1638348095621
  },
  "priceTagInfo": {
    "barCodeList": [],
    "oemModelList": [
      "ZKC75V-DN"
    ],
    "state": 0,
    "batteryLevel": 5
  },
  "displayInfo": {
    "needPriceTagInfo": true,
    "needBindInfo": true,
    "needItemInfo": false,
    "needRelationalInfo": false
  },
  "itemQueryCondition": {
    "barCode": "",
    "itemTitle": "cola330ml"
  }
}
```

```
},  
"pageSize": 10,  
"currentPage": 1  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 3 12 13 70)

Example 1:

```
{  
  "success": false,  
  "code": 19015,  
  "message": "No store matches the external store ID",  
  "data": [  
    "3",  
    "4"  
  ],  
  "translate": false  
}
```

Example 2:

```
{  
  "success": false,  
  "code": 15058,  
  "message": "ESL model does not exist",  
  "data": [  
    "7758258"  
  ],  
  "translate": false  
}
```

Example 3: Display ESL information and binding relationship

```
{  
  "success": true,  
  "code": 10000,  
  "message": "Operation successful",  
  "data": {  
    "pageReturn": {  
      "list": [  

```

```
{
  "id": 80,
  "bindType": 1,
  "templateId": null,
  "bindTime": null,
  "updateTime": null,
  "priceTagCode": "100121669",
  "model": null,
  "softVersion": "",
  "state": 0,
  "battery": 84,
  "batteryLevel": 5,
  "lastCommuTime": "2021-11-02 08:27:54",
  "apSignal": null,
  "shelfNo": null,
  "deviceType": 1,
  "turnOver": 0,
  "temperature": null,
  "noData": null,
  "apMac": "A0A3B80E4380",
  "apList": null,
  "itemBindInfos": [
    {
      "id": 15,
      "itemOrder": null,
      "itemBarCode": "111",
      "itemTitle": "cola330ml",
      "templateAttr": "default"
    }
  ],
  "storeId": 1632276202577,
  "storeName": null,
  "externalStoreId": null
},
{
  "totalElements": 14
},
{
  "itemInfo": []
},
{
  "translate": false
}
```

Example 4: Display ESL information, binding relationship, and product information

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "pageReturn": {
      "list": [
        {
```

```
"id": 80,
"bindType": 1,
"templateId": null,
"bindTime": null,
"updateTime": null,
"priceTagCode": "100121669",
"model": null,
"softVersion": "",
"state": 0,
"battery": 84,
"batteryLevel": 5,
"lastCommuTime": "2021-11-02 08:27:54",
"apSignal": null,
"shelfNo": null,
"deviceType": 1,
"turnOver": 0,
"temperature": null,
"noData": null,
"apMac": "A0A3B80E4380",
"apList": null,
"itemBindInfos": [
{
"id": 15,
"itemOrder": null,
"itemBarCode": "111",
"itemTitle": "cola330ml",
"templateAttr": "default"
}
],
"storeId": 1632276202577,
"storeName": null,
"externalStoreId": null
},
"totalElements": 14
},
"itemInfo": [
{
"id": 15,
"type": 1,
"price": 1,
"spec": "Specification",
"productCode": "8521",
"barCode": "111",
"agencyId": 1597317958637,
"merchantId": 1632276202525,
"storeId": 0,
"originalPrice": 3.35,
"memberPrice": 2.35,
"productArea": "China",
"classLevel": "level",
```

```
"attrCategory": "default",
"attrName": "default",
"sourceCode": null,
"proStartTime": "2018-10-10 10:00:00",
"proEndTime": "2018-10-10 10:00:00",
"number": null,
"unit": "Unit",
"productSku": "123",
"createdTime": "2022-02-16 16:49:36",
"itemTitle": "cola330ml",
"shortTitle": "cola",
"brandName": null,
"suggestPrice": null,
"qrCode": "",
"promotionText": "discount",
"coverPic": null,
"custFeature1": "11",
"custFeature2": "2",
...
"custFeature50": "50",
"updateTime": "2022-03-29 14:57:05",
"delTime": null,
"stock1": 11,
"stock2": 22,
"stock3": 33,
"isSpecs": 0,
"nfcUrl": "nfc",
"storeName": null,
"label": null
}
],
},
"translate": false
}
```

// Display ESL information, binding relationship, product information, and store information

```
{
"success": true,
"code": 10000,
"message": "Operation successful",
"data": {
"pageReturn": {
"list": [
{
"id": 80,
"bindType": 1,
"templateId": null,
"bindTime": null,
"updateTime": null,
```

```
"priceTagCode": "100121669",
"model": null,
"softVersion": "",
"state": 0,
"battery": 84,
"batteryLevel": 5,
"lastCommuTime": "2021-11-02 08:27:54",
"apSignal": null,
"shelfNo": null,
"deviceType": 1,
"turnOver": 0,
"temperature": null,
"noData": null,
"apMac": "A0A3B80E4380",
"apList": null,
"itemBindInfos": [
{
"id": 15,
"itemOrder": null,
"itemBarCode": "111",
"itemTitle": "cola330ml",
"templateAttr": "default"
}
],
"storeId": 1632276202577,
"storeName": "default",
"externalStoreId": null
},
"totalElements": 14
},
"itemInfo": [
{
"id": 15,
"type": 1,
"price": 1,
"spec": "Specification",
"productCode": "8521",
"barCode": "111",
"agencyId": 1597317958637,
"merchantId": 1632276202525,
"storeId": 0,
"originalPrice": 3.35,
"memberPrice": 2.35,
"productArea": "China",
"classLevel": "level",
"attrCategory": "default",
"attrName": "default",
"sourceCode": null,
"proStartTime": "2018-10-10 10:00:00",
"proEndTime": "2018-10-10 10:00:00",
```

```
"number": null,
"unit": "Unit",
"productSku": "123",
"createTime": "2022-02-16 16:49:36",
"itemTitle": "cola330ml",
"shortTitle": "cola",
"brandName": null,
"suggestPrice": null,
"qrCode": "",
"promotionText": "discount",
"coverPic": null,
"custFeature1": "11",
"custFeature2": "2",
...
"custFeature50": "50",
"updateTime": "2022-03-29 14:57:05",
"delTime": null,
"stock1": 11,
"stock2": 22,
"stock3": 33,
"isSpecs": 0,
"nfcUrl": "nfc",
"storeName": null,
"label": null
}
],
},
"translate": false
}
```

4.23 Set Merchant Response Callback URL for Picking ESLs

API Definition

URL	http://[server address]/zk/config/settingCallbackUrl
Example	http://esl.zkong.com/zk/config/settingCallbackUrl
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Set the callback URL for the merchant.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
url	body	String	true	response URL
fromAppKey	body	String	false	callerkey
fromCode	body	String	false	Caller code
toAppKey	body	String	false	receiverkey
toAppKeyCode	body	String	false	receivercode
api	body	String	false	Request authentication API
secret	body	String	false	Carry secret key
callbackType	body	String	true	callbacktype, support simple,cainiao where simple is a normal response

Request Example

```
{
  "url": "https://daily-fortune.wt.cainiao.com/gateway/link.do",
  "fromAppKey": "dell_client",
  "fromCode": "device1",
  "toAppKey": "cainiao_iot_fortune",
  "toCode": "cainiao_iot_fortune",
  "api": "UPDATE_EINK",
  "secret": "!TGB6yhn7dell8c,",
  "callbackType": "simple"
}
```

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	Message

Response Data	data	Json	Configured callback information
---------------	------	------	---------------------------------

Response Example 1

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "url": "https://daily-fortune.wt.cainiao.com/gateway/link.do",
    "fromAppKey": "dell_client",
    "fromCode": "device1",
    "toAppKey": "cainiao_iot_fortune",
    "toCode": "cainiao_iot_fortune",
    "api": "UPDATE_EINK",
    "secret": "!TGB6yhn7dell8c,",
    "callbackType": "simple"
  },
  "translate": false
}
```

Response Example 2

```
{
  "success": false,
  "code": 10009,
  "data": null,
  "message": " param invalid"
}
```

4.23.1 Light-off Report Callback Parameters

Callback Request Parameters

Parameter Name	Parameter Type	Data Type	Notes
barCode	body	String	ESL Barcode
mac	body	String	ESL MAC
callType	body	String	Callback type 1 light-off callback, 2 long-press/short-press callback

merchantId	body	Integer	merchantnumber
itemBarCode	body	String	Product Code
itemName	body	String	Product Name
serialNum	body	Integer	Closed light index 0 first light, 1 second light
type	body	Integer	flashtype1startflash0offflash

Example

```
{
  "barCode": "A0A3B84A2929",
  "mac": "A0A3B84A2929",
  "callType": 1,
  "merchantId": 1629967515267,
  "itemBarCode": "1000003",
  "itemName": "Newly Listed Product",
  "serialNum": 0,
  "type": 1
}
```

4.23.2 Short-press/Long-press Report Callback Parameters

Callback Request Parameters

Parameter Name	Parameter Type	Data Type	Notes
barCode	body	String	ESL Barcode
mac	body	String	ESL MAC
callType	body	Integer	Callback type 1 light-off callback, 2 long-press/short-press callback
pageNum	body	Integer	Page Number starting from 0
ledData	body	List<object>	Short-press/long-press callback data
serialNum	body	Integer	LED index, starting from 0

	type	body	Integer	Type: 0 reserved, 1 short press, 2 long press
--	------	------	---------	---

Example

```
{
  "barCode": "A0A3B84A2929",
  "mac": "A0A3B84A2929",
  "callType": 2,
  "pageNum": 0,
  "ledData": [
    {
      "serialNum": 0,
      "type": 1
    },
    {
      "serialNum": 1,
      "type": 2
    }
  ]
}
```

4.24 Query ESL Barcode by ESL MAC or UUID

API Definition

URL	http://[server address]/zk/pricetag/getPricetagBarcodeByPricetagMac
Example	http://esl.zkong.com/zk/pricetag/getPricetagBarcodeByPricetagMac
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query the ESL barcode by ESL MAC or UUID. Either uuid or mac is required. If both are provided, uuid takes priority.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
pricetagMac	body	String	false	ESL MAC
pricetagUuid	body	String	false	ESL UUID

Request Example

<http://esl.zkong.com/zk/pricetag/getPricetagBarcodeByPricetagMac?pricetagUuid=1D6004338E0000>

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	Message
Response Data	data	String	ESL Barcode

Response Example 1

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": "807853349",
  "translate": true,
  "originalMessage": null
}
```

Response Example 2

```
{
  "success": false,
  "code": 15069,
  "message": "No ESL was found. Please contact the manufacturer to import the key file.",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.25 ESL Management - Export Binding Relationships

API Definition

URL	http://[server address]/zk/bind/bindExport/cn
Example	http://esl.zkong.com/zk/bind/bindExport/cn

Request Method	POST
Content Type	application/json;charset=utf-8
Description	Export the binding relationships between ESLs and products.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
storeIds	body	List<Long>	false	Store ID list; if not provided, all stores of the current user are used
model	body	String(20)	false	ESL Model (if not provided, query all)
state	body	Integer(1)	false	ESL status (0: offline, 1: online)
deviceType	body	Integer(1)	false	Color (0: black/white, 1: black/white/red, 2: black/white/yellow, 3: multi-color)
batteryLevel	body	Integer(1)	false	ESL battery level (1: urgently needs charging, 2: low battery, 3: medium battery, 4: high battery, 5: full battery)
turnOver	body	Integer(1)	false	Whether inverted (0: no, 1: yes)
apSignalStart	body	Integer	false	Minimum Signal Value
apSignalEnd	body	Integer	false	Maximum Signal Value
priceTagCode	body	String(60)	false	ESL Barcode
itemBarCode	body	String(100)	false	Product Code

itemTitle	body	String(100)	false	Product Name
shelfNo	body	String(60)	false	Shelf Number
labelList	body	List<String>	false	Label List
orgIdList	body	List<Integer>	false	Organization ID List

Request Example

```

Request URL:
http://esl.zkong.com/zk/bind/bindExport/cn
Request Body:
{
  "model": "ZKC21V-N",
  "state": "1",
  "deviceType": null,
  "batteryLevel": null,
  "turnOver": "0",
  "apMacList": [],
  "storeIds": [
    1740556397882
  ],
  "orgIdList": [
    105
  ],
  "labelList": [
    "Midea",
    "Little Swan"
  ],
  "priceTagCode": null,
  "itemBarCode": null,
  "itemTitle": null,
  "shelfNo": null,
  "apSignalStart": "2",
  "apSignalEnd": "99"
}

```

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	Message
Response Data	data	OutputStream	File Stream

Response Example 1

```
{
  "success": false,
  "code": 19018,
  "message": "Store does not exist or is disabled",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

Response Example 2

```
{
  "success": false,
  "code": 10030,
  "message": "You do not have permission to access this",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

4.26 Query ESL Status and Related Business Information

API Definition

URL	http://[server address]/zk/erp/esl/adminBusinessInfoList
Example	http://esl.zkong.com/zk/erp/esl/adminBusinessInfoList
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query ESL status and related business information. If storeId and externalStoreId are both used in the query, externalStoreId takes priority.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
deviceType	body	Integer	false	Device Type1: ESL 2: AP Default: 1

storeId	body	Long	false	Store ID
externalStoreId	body	String	false	External Store ID
merchantId	body	Long	Notes	Merchant ID, required when querying data using external store ID
pageSize	body	Integer	false	Page Size Default: 10
currentPage	body	Integer	false	Page Number, Default: 1

Request Example

Request URL:

<http://esl.zkong.com/zk/erp/esl/adminBusinessInfoList>

Request Body:

```
{
  "deviceType": 1,
  "externalStoreId": "0815",
  "merchantId": 1644307350261,
  "pageSize": 10,
  "currentPage": 1
}
```

Response Parameters (ESL Query):

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	
Response Data	data	Json	
Total Records	totalElements	Integer	
Current Page	currentPage	Integer	

Page Size		pageSize	Integer	
Total Pages		totalPage	Integer	
ESL Information List		list	List	
	ESL ID	id	Integer	ESL Auto-increment ID
	ESL Barcode	barCode	String	
	ESL MAC	mac	String	
	AP MAC	apMac	String	Last Communication AP
	Width	width	Integer	Width Resolution
	Height	height	Integer	Height Resolution
	Device Type (Supported Colors)	deviceType	Integer	0: black/white, 1: black/white/red, 2: black/white/yellow, 4: seven-color type 1, 5: black/white/red/yellow, 6: seven-color type 2
	Device Model	model	String	
	Store ID	storeId	Long	

	External Store ID	externalStoreId	String	
	Merchant ID	merchantId	Long	
	Agency ID	agencyId	Long	
	Binding Status	bindState	Integer	0: unbound 1: bound
	Online Status	state	Integer	0: offline 1: online
	Last Upgrade Time	upgradeTime	String	
	Added Time	addTime	Integer	
	Upgrade Status	upgradeState	Integer	0: issuing 1: issuing failed 2: upgrading 3: upgrade successful 5: upgrade failed
	Whether inverted	turnOver	Integer	0: no 1: yes
	Reboot Status	rebootStatus	Integer	1: issuing 2: issued successfully 3: issuing failed
	Remaining Battery	battery	Integer	Range: 1-100
	Battery Level	batteryLevel	Integer	Range: 1-5
	Base Station Signal	apSignal	Integer	

	AP list near the ESL	apList	String	
	Binding Status	bindState	Integer	0: unbound, 1: bound
	Binding Time	bindTime	String	
	Maximum Signal Strength	apSignal	Integer	
	ESL Time	currentTime	String	Unit: seconds
	Last Communicatio n Time	lastCommunicationTi me	String	
	Software Version	softVersion	String	
	Hardware Version	hardVersion	String	
	Image Capacity	picNum	Integer	
	Chip Type	chipSolution	String	
	NFC UUID	uuid	String	
	Last Reporting AP	lastReportApMac	String	
	Voltage	voltage	Double	
	Default Sleep Time	defaultSleepTime	Integer	
	Actual Sleep Time	actualSleepTime	Integer	
	Roaming Mode	wanderMode	Integer	1: normal 2: roaming
	Base AP MAC	baseApMac	String	

	Product Code	itemBarCode	String	
	Product Name	itemTitle	String	
	AP Status	apState	Integer	
	Floor	floor	Integer	
	Positioning Time	locationTime	String	
	Shelf Number	shelfNo	String	
	Shelf Type	shelfType	String	

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "pageSize": 10,
    "currentPage": 1,
    "totalPage": 2,
    "totalElements": 15,
    "list": [
      {
        "id": 629,
        "barCode": "863800777",
        "mac": "A0A3108573E4",
        "apMac": "A0A3B815F040",
        "width": 480,
        "height": 648,
        "deviceType": 6,
        "model": "ZKC58B-NEA7",
        "agencyId": 1597317958637,
        "merchantId": 1644307350261,
        "storeId": 1723685930724,
        "bindState": 1,
        "state": 1,
        "upgradeTime": "2024-12-04 14:23:11",
        "addTime": "2024-12-09 17:30:20",
        "upgradeState": 4,
        "turnOver": 0,
        "rebootStatus": 0,

```

```

"creator": 99999,
"createTime": "2024-12-03 14:43:34",
"updater": 77,
"updateTime": "2024-12-09 14:09:09",
"battery": 100,
"batteryLevel": 5,
"sequence": 0,
"apSignal": 83,
"apList": "{\\\"A0A3B8119F80\\\":75,\\\"A0A3B815F040\\\":83,\\\"A0A3B81196E0\\\":73,\\\"A0A3B8167850\\\":76,\\\"A0A3B9643670\\\":72,\\\"A0A3B8167AF0\\\":72}",
"currentTime": 443182,
"currentSystemTime": 1733736620,
"abilities": "0",
"lastCommunicationTime": "2024-12-09 17:30:20",
"softVersion": "3.4.4",
"hardVersion": "5331",
"picNum": 11,
"chipSolution": 2,
"uuid": "04C3137E250269",
"lastReportApMac": "A0A3B815F040",
"voltage": 3.1,
"defaultSleepTime": 30,
"actualSleepTime": 30,
"wanderMode": "1",
"productType": 0,
"baseApMac": "A0A3B815F040",
"itemBarCode": "58",
"itemTitle": "5.8-inch ESL Test Verification",
"apState": 1,
"floor": 5,
"locationTime": "2024-12-09 17:35:10",
"shelfNo": "102",
"shelfType": "2"
}
],
},
"translate": true,
"originalMessage": null
}

```

Response Parameters (AP Base Station Query):

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details

Message	message	String	
Response Data	data	Json	
Total Records	totalElements	Integer	
Current Page	currentPage	Integer	
Page Size	pageSize	Integer	
Total Pages	totalPage	Integer	
AP Information List	list	List	
APID	id	Integer	AP Auto-increment ID
AP Name	apName	String	
AP MAC	mac	String	
AP Model	model	String	
Online Status	online	Integer	0: offline, 1: online
Last Upgrade Status	upgradeState	Integer	
Last Upgrade Time	upgradeTime	Date	"
Last Online Time	joinTime	Date	
Last Offline Time	offlineTime	Date	
External Store ID	externalStoreId	String	
Merchant ID	merchantId	Long	
Store ID	storeId	Long	

Number of Connected ESLs	eslCount	Integer	
Software Version	softVersion	String	
Hardware Version Time	hardVersion	String	
IP Address	ip	String	
Reboot Status	rebootState	Integer	0: successful 1: issuing 2: failed
Reboot Time	rebootTime	Date	
2G Wi-Fi Name	wifiName2G	String	
5G Wi-Fi Name	wifiName5G	String	

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "pageSize": 10,
    "currentPage": 1,
    "totalPage": 1,
    "totalElements": 5,
    "list": [
      {
        "id": 158,
        "mac": "A0A3B815F040",
        "apName": "A0A3B815F040",
        "merchantId": 1644307350261,
        "storeId": 1723685930724,
        "model": "ZAP-CML",
        "online": 1,
        "comment": ""
      }
    ]
  }
}
```

```
"upgradeState": null,
"upgradeTime": null,
"joinTime": "2024-12-12T18:11:58.000+00:00",
"offlineTime": "2024-12-12T14:11:12.000+00:00",
"eslCount": 12,
"statisticTime": null,
"wordModePlatformState": 0,
"createUserId": 60,
"createTime": "2024-12-10T08:58:54.000+00:00",
"updateUserId": 60,
"updateTime": "2024-12-16T08:10:30.000+00:00",
"deleted": 0,
"softVersion": "6.2.006_Beta1115",
"hardVersion": "5",
"ip": "192.168.16.42",
"cpu": null,
"memory": null,
"onlineTime": 0,
"reportWorkModeReportState": null,
"reportContentJson": null,
"reportTime": null,
"rebootState": 0,
"rebootTime": "2024-12-10T08:59:13.000+00:00",
"wifiName2G": null,
"wifiName5G": null,
"channel2G": 0,
"channel5G": 0,
"wifiStatus": null,
"wifiSetTime": null,
"yawAngle": 118.9,
"pitchAngle": 1.3,
"rollAngle": -0.7,
"externalStoreId": "0815"
}
],
},
"translate": true,
"originalMessage": null
}
```

4.27 Refresh Image Using Specified Template

API Definition

URL	http://[server address]/zk/erp/esl/changeTemplateForTags
Example	http://esl.zkong.com/zk/erp/esl/changeTemplateForTags
Request Method	POST

Content Type	application/json;charset=utf-8
Description	Refresh images for specified ESLs using a specified template.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
barCodes	query	List<String>	true	ESLbarCode
merchantId	query	Long	true	Merchant ID
storeId	body	Long	true	Store ID
externalStoreId	body	String (40)	false	External Store ID
useExternalStoreId	body	Integer	false	Use externalStoreId 1 - use, 0 - do not use; if not provided, do not use by default
templateAttrCategory	body	String(120)	true	Template Category
templateAttr	body	String(120)	true	Template Attribute

Request Example

Request URL:

http://esl.zkong.com/zk/erp/esl/changeTemplateForTags

Request Body:

```
{
  "storeId": 1588041156676,
  "merchantId": 147896541231,
  "barCodes": ["A1A2A4A3A4", "A1A2A4A3A5"],
  "templateAttrCategory": "test",
  "templateAttr": "test",
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request	Boolean	true/false

	Successful		
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 49)

```
{
  "success":true,
  "code":10000,
  "data": "",
  "message": "Success",
  "translate": true,
  "originalMessage": "Success"
}
```

4.28 Refresh Image Using Default Template

API Definition

URL	http://[server address]/zk/erp/esl/restoreTemplateForTags
Example	http://esl.zkong.com/zk/erp/esl/restoreTemplateForTags
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Refresh images for specified ESLs using a specified template.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
barCodes	body	List<String>	true	ESLbarCode
merchantId	body	Long	true	Merchant ID

storeId	body	Long	true	Store ID
storeId	body	Long	true	Store ID
externalStoreId	body	String (40)	false	External Store ID

Request Example

Request URL:

<http://es1.zkong.com/zk/erp/es1/restoreTemplateForTags>

Request Body:

```
{
  "storeId": 1588041156676,
  "merchantId": 147896541231,
  "barCode": ["A1A2A4A3A4", "A1A2A4A3A5"]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 49)

```
{
  "success": true,
  "code": 10000,
  "data": "",
  "message": "Success",
  "translate": true,
  "originalMessage": "Success"
}
```

4.29 Get ESL Bound Shelf Information

API Definition

URL	http://[server address]/zk/erp/esl/getBindShelfNum
Example	http://esl.zkong.com/zk/erp/esl/getBindShelfNum
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query shelf information by barcode and return the shelf number, ESL barcode, and product information.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
barcode	query	String(60)	true	ESL Barcode

Request Example

Request URL:
<http://esl.zkong.com/zk/bind/erp/esl/getBindShelfNum?barcode=8002087682>
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 41)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
```

```
        "barcode": "800208768",
        "shelfNum": "112.113",
        "itemBindInfos": [
            {
                "id": null,
                "itemOrder": 1,
                "itemBarCode": "1",
                "itemTitle": "cola330ml",
                "templateAttr": "default"
            }
        ],
        "translate": true,
        "originalMessage": null
    }
```

Example 2 (usually the provided barcode is not bound or does not exist)

```
{
  "success": false,
  "code": 10009,
  "message": "Parameter error",
  "data": null,
  "translate": true,
  "originalMessage": "param invalid"
}
```

4.30 PTL Button Information Push (provided by the user system)

API Definition

URL	usercustom
Example	http://esl.notice.com/pressButton/notice
Request Method	POST
Content Type	application/json;charset=utf-8
Description	This API must be provided by the user system. When a PTL button is pressed, the picking system pushes button information to this API provided by the user system. The request parameters must follow the format defined in this document. No specific response format is required.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
barCode	body	String	true	PTL Barcode
mac	body	String	true	PTL MAC Address
pageNum	body	Integer	true	Image page number when the button is pressed
ledData	body	List<Object>	true	Button Event List
type	body	Integer	true	Button type (1: short press, 2: long press)
serialNum	body	Integer	true	LED index (0: left key, 1: right key, 2: main key)

Request Example

Request URL:
<http://esl.notice.com/pressButton/notice>

Request Body:

```
{
  "barCode": "80060777",
  "mac": "A0A3B8273D77",
  "pageNum": 1,
  "ledData": [
    {
      "type": 1,
      "serialNum": 2
    }
  ]
}
```

4.31 Execute PTL Loop Flashing

API Definition

URL	http://{server address}/zk/loopFlash/execute/v2
Example	http://esl.zkong.com/zk/loopFlash/execute/v2
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Execute loop flashing on PTL devices.

Request Parameters

Parameter Name		Parameter Type	Data Type	Required	Notes
Authorization		header	String	true	Authorization token
useExternalStoreId		query	Integer (1)	false	Whether to use externalStoreId (0: no, 1: yes)
externalStoreId		query	String (100)	false	External store ID (required when used)
storeId		body	Long	true	Store ID
ledList		body	List<Object>	true	Flashing List
	barcode	body	String (100)	true	PTL
	lightTotalTime	body	Integer (5)	false	Flashing duration (seconds), not exceeding 14,399 seconds; default: 600 seconds
	lightFrequency	body	Integer (1)	false	Flashing frequency (1: slow flash, 2: fast flash), default: slow flash
	lightColors	body	List<Integer>	true	Loop flashing color group. PTL color values can be 1-8, corresponding to red, green, blue, yellow, orange, cyan, purple, and white.

Request Example

Request URL:
<http://esl.zkong.com/zk/loopFlash/execute>
 Request Body:

```
{
  "storeId":1630484444737,
```

```
"ledList": [
{
"barcode": "80060777",
"lightTotalTime": 3600,
"lightFrequency": 1,
"lightColors": [
1,
2,
5
]
}
]
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 41 45 96 97 98)

```
{
"success":true,
"code": 10000,
"data": null,
"message": "Operation successful",
"translate": true,
"originalMessage": null
}
```

4.32 Stop PTL Loop Flashing

API Definition

URL	http://{server address}/zk/loopFlash/stop
Example	http://esl.zkong.com/zk/loopFlash/stop
Request Method	POST

Content Type	application/json;charset=utf-8
Description	Stop loop flashing on PTL devices.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer (1)	false	Whether to use externalStoreId (0: no, 1: yes)
externalStoreId	query	String (100)	false	External store ID (required when used)
storeId	body	Long	true	Store ID
labelBarcodes	body	List<String>	true	PTL Barcode List

Request Example

Request URL:
<http://esl.zkong.com/zk/loopFlash/stop>
 Request Body:

```
{
  "storeId":1630484444737,
  "labelBarcodes": [
    "80060777"
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 41 45 96 97 98)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

5. Template Import and Query

5.1 Import ESL Template

API Definition

URL	http://[server address]/zk/template/importTemplateRefactorFile/{sceneNumber}
Example	http://esl.zkong.com/zk/template/importTemplateRefactorFile/1
Request Method	POST
Content Type	multipart/form-data
Description	Import ESL templates for different scenarios to a merchant/store through the HTTP API.

Request Parameters

Variable Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
merchantId	query	Long(20)	true	Merchant ID
storeId	query	Long(20)	false	Store ID If left blank, import to merchant level
sceneNumber	path	Integer(1)	true	number 1:single-productbindtemplate 2:unbindtemplate

				3:segment screentemplate 4:multi-productbindtemplate
file	body	file	true	Template File Supports a single file (maximum 50 KB) and a compressed package (ZIP format, maximum 500 KB)

Request Example

Request URL:
<http://es1.zkong.com/zk/template/importTemplateRefactorFile/1?merchantId=1585812365204&storeId=1632276202577>
Request Body:

```
{
  "multipartFile": "<uploadfile>"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 41 71 72 73 74 75 76)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "success",
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 12042,
  "data": null,
  "message": "The template file format is incorrect",
  "translate": true,
  "originalMessage": null
}
```

5.2 Delete ESL Template

API Definition

URL	http://[server address]/zk/template/delete/{id}/{isCompel}
Example	http://esl.zkong.com/zk/template/delete/123/1
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Delete ESL templates from the ESL platform through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Template ID
isCompel	path	Integer	true	Force Delete 0: no, any other value: yes

Request Example

Request URL:
<http://esl.zkong.com/zk/template/delete/123/1>
Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false

code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 77 78)

```
{
  "success": true,
  "code": 10000,
  "data": null,
  "message": "Operation successful",
  "translate": true,
  "originalMessage": null
}
```

5.3 Import Merchant Template to Store

API Definition

URL	http://[server address]/zk/template/copyMerchantTo/{storeId}
Example	http://esl.zkong.com/zk/template/copyMerchantTo/1632276202577
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Copy merchant templates to a store.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
storeId	path	Integer(20)	true	Store ID
ids	body	List<String>	true	templateidlist

Request Example

Request URL:
<http://esl.zkong.com/zk/template/copyMerchantTo/1632276202577>

Request Body:

[11, 22]

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 79)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.4 Add Template Category

API Definition

URL	http://[server address]/zk/attrCategory/add
Example	http://esl.zkong.com/zk/attrCategory/add
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Add Template Category

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

categoryName	body	String(120)	true	Template Category Name
--------------	------	-------------	------	------------------------

Request Example

Request URL:
<http://esl.zkong.com/zk/attrCategory/add>
 Request Body:

```
{
  "categoryName": "test"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Null	

Response Example (related error code no.: 80 81)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.5 Query Merchant Template Categories

API Definition

URL	http://[server address]/zk/attrCategory/findList
Example	http://esl.zkong.com/zk/attrCategory/findList
Request Method	GET
Content Type	application/json;charset=utf-8

Description	Query complete template category information for the merchant corresponding to this token.
--------------------	--

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

Request Example

Request URL:
<http://es1.zkong.com/zk/attrCategory/findList>
 Request Body:
 None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.):

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": [
    {
      "id": 12,
      "categoryName": "default",
      "agencyId": 1597317958637,
      "merchantId": 1632276202525,
      "storeId": null,
      "addTime": "2021-09-22 10:03:22",
      "delTime": null,
      "updateTime": null,
      "operateUser": "jsk123",
    }
  ]
}
```

```

"flag": 1
},
{
  "id": 13,
  "categoryName": "3",
  "agencyId": 1597317958637,
  "merchantId": 1632276202525,
  "storeId": null,
  "addTime": "2021-10-19 20:21:17",
  "delTime": null,
  "updateTime": null,
  "operateUser": "jsk123",
  "flag": 1
},
{
  "id": 123,
  "categoryName": "Normal",
  "agencyId": 1597317958637,
  "merchantId": 1632276202525,
  "storeId": null,
  "addTime": "2022-06-29 09:58:06",
  "delTime": null,
  "updateTime": null,
  "operateUser": "jsk1234",
  "flag": 1
}
],
"translate": true,
"originalMessage": null
}

```

5.6 Delete Template Category

API Definition

URL	http://[server address]/zk/attrCategory/del/{id}
Example	http://esl.zkong.com/zk/attrCategory/del/159
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Delete ESL template categories from the ESL platform through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
----------------	----------------	-----------	----------	-------

Authorization	header	String	true	Authorization token
id	path	Integer	true	Template ID

Request Example

Request URL:
<http://esl.zkong.com/zk/attrCategory/del/159>
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 82 83)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.7 Add Template Attribute

API Definition

URL	http://[server address]/zk/templateAttr/add
Example	http://esl.zkong.com/zk/templateAttr/add
Request Method	POST
Content Type	application/json;charset=utf-8

Description	Add Template Attribute
--------------------	------------------------

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
categoryId	body	String	true	Template Category ID
categoryName	body	String(120)	true	Template Category Name
attrName	body	String(120)	true	Template Attribute Name

Request Example

Request URL:
<http://es1.zkong.com/zk/templateAttr/add>
 Request Body:

```
{
  "categoryId": 123,
  "categoryName": "Normal",
  "attrName": "test"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 81 84 85)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
}
```

```
"originalMessage": null
}
```

5.8 Query Merchant Template Attributes

API Definition

URL	http://[server address]/zk/templateAttr/findList
Example	http://esl.zkong.com/zk/templateAttr/findList
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query complete template attribute information for the merchant corresponding to this token.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

Request Example

Request URL:
<http://esl.zkong.com/zk/templateAttr/findList>
 Request Body:

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.):

```
{
  "success": true,
```

```
"code": 10000,
"message": "Operation successful",
"data": [
{
  "id": 1632276202598,
  "categoryId": 12,
  "categoryName": "default",
  "attrCode": null,
  "attrName": "default",
  "agencyId": 1597317958637,
  "merchantId": 1632276202525,
  "storeId": null,
  "isPublic": null,
  "sort": null,
  "addTime": "2021-09-22 10:03:22",
  "delTime": null,
  "updateTime": null,
  "operateUser": "jsk123",
  "flag": 1
},
{
  "id": 1634646072425,
  "categoryId": 12,
  "categoryName": "default",
  "attrCode": null,
  "attrName": "1",
  "agencyId": 1597317958637,
  "merchantId": 1632276202525,
  "storeId": null,
  "isPublic": null,
  "sort": null,
  "addTime": "2021-10-19 20:21:12",
  "delTime": null,
  "updateTime": null,
  "operateUser": "jsk123",
  "flag": 1
}
],
"translate": true,
"originalMessage": null
}
```

5.9 Delete Template Attribute

API Definition

URL	http://[server address]/zk/templateAttr/del/{id}
-----	--

Example	http://esl.zkong.com/zk/templateAttr/del/1670240463031
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Delete ESL template attributes from the ESL platform through the HTTP API.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Long	true	Template Attribute ID

Request Example

Request URL:
<http://esl.zkong.com/zk/templateAttr/del/1670240463031>
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 86)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.10 Query Complete ESL Model Information

API Definition

URL	http://[server address]/zk/pricetagModel/getAll
Example	http://esl.zkong.com/zk/pricetagModel/getAll
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query all complete information related to ESL models, including width, height, resolution, model, size, etc.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

Request Example

Request URL:
http://esl.zkong.com/zk/pricetagModel/getAll
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.):

```
{
  "success": true,
  "code": 10000,
```

```
"message": "Operation successful",
"data": [
{
  "id": 65,
  "size": "1.54",
  "model": "ZKC15S",
  "oemModel": "ZKC15S_ZKK",
  "resolution": "152*152",
  "width": 152,
  "height": 152,
  "color": 1,
  "modelSpec": null,
  "temperature": 1,
  "weight": null,
  "battery": null,
  "hardVersion": null,
  "isHighResolution": null,
  "createdTime": "2019-05-12 10:33:12",
  "imgPath": null,
  "screenType": 0,
  "nfc": 1,
  "flag": 1,
  "isDefaultReverse": 0
},
{
  "id": 101,
  "size": "1.54",
  "model": "ZKC15S-N",
  "oemModel": "ZKC15S-N",
  "resolution": "152*152",
  "width": 152,
  "height": 152,
  "color": 1,
  "modelSpec": null,
  "temperature": 1,
  "weight": null,
  "battery": null,
  "hardVersion": "1.0",
  "isHighResolution": null,
  "createdTime": "2019-12-06 11:03:33",
  "imgPath": null,
  "screenType": 0,
  "nfc": 1,
  "flag": 1,
  "isDefaultReverse": 0
}
],
"translate": true,
"originalMessage": null
}
```

5.11 Export ESL Template

API Definition

URL	http://[server address]/zk/template/exportTemplateRefactorFile/{sceneNumber}
Example	http://esl.zkong.com/zk/template/exportTemplateRefactorFile/{sceneNumber}
Request Method	POST
Content Type	multipart/form-data
Description	Export templates as a ZIP file through the HTTP API.

Request Parameters

Variable Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
templateBaselds	body	List<Long> (100)	true	templateIDlist
sceneNumber	path	Integer (1)	false	number 1:single-productbindtemplate 2:unbindtemplate 3:segment screentemplate 4:multi-productbindtemplate

Request Example

Request URL:
<http://esl.zkong.com/zk/template/exportTemplateRefactorFile/1>
 Request Body:
 [1,15,20]

Response Parameters

Variable Name	Parameter Name	Data Type	Description
---------------	----------------	-----------	-------------

<Output Stream>	<Output Stream>	OutputStream	Output a.zip file for download
-----------------	-----------------	--------------	--------------------------------

Response Example (related error code no.: 87)

```
{
  "success": true,
  "code": 12045,
  "message": "No template selected",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.12 Enable ESL Template

API Definition

URL	http://[server address]/zk/template/enable/{id}/{isCompel}
Example	http://esl.zkong.com/zk/template/enable/808/1
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Enable the ESL template with the specified number. When force enable is used, any other enabled template of the same type will be automatically disabled. When force enable is not used, if another template of the same type is already enabled, the system will prompt that a template of the same type is already enabled.

Request Parameters

Variable Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Template Number
isCompel	path	Integer(1)	true	Force enable 0: no, any other value: yes

Request Example

Request URL:
<http://esl.zkong.com/zk/template/enable/808/1>

Request Body:

None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 16 88)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 12024,
  "message": "A template of the same type is already enabled. Disable that template and enable this one?",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

5.13 Disable ESL Template

API Definition

URL	http://[server address]/zk/template/disable/{id}/{isCompel}
Example	http://esl.zkong.com/zk/template/disable/808/1
Request Method	GET
Content Type	application/json;charset=utf-8

Description	Disable the ESL template with the specified number. When force disable is used, the template is disabled even if it is currently in use. When force disable is not used, if the template is currently in use, the system will prompt that ESLs are using this template and ask whether to disable it.
--------------------	---

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	body	Integer	true	Template Number
isCompel	body	Integer (1)	true	Force enable 0: no, any other value: yes

Request Example

Request URL:
<http://esl.zkong.com/zk/template/disable/808/1>
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 16 89)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success":false,
  "code": 12023,
  "message": "Some ESLs are using this template. Disable this template?",
  "data":null,
  "translate": true,
  "originalMessage": null
}
```

5.14 Query Store Templates

API Definition

URL	http://[server address]/zk/template/list
Example	http://esl.zkong.com/zk/template/list
Request Method	POST
Content Type	application/json;charset=utf-8
Token	Add Authorization: [token value returned by login API 3.2] to the request header.
Description	Query ESL templates under a store.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
page	path	Integer	true	Page number (default first page number is 0)
size	path	Integer	true	Page Size
useExternalStoreId	query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String (40)	false	External Store ID required when used
Object	body	Object	true	Query condition object
storeId	body	Integer(20)	true	Store ID

hardwareType	body	String(1)	false	Supported Colors "0": black/white, "1": black/white/red "2": black/white/yellow, "3": multi-color
sceneNumber	body	Integer(1)	false	Template Type 1:single- producttemplate 2:unbindtemplate 3:segment screentemplate 4:multi- producttemplate
size	body	String(1)	false	Template Size

Request Example

Request URL:
<http://es1.zkong.com/zk/template/list/0/10>
 Request Body:

```
{
  "storeId": 1584608946868,
  "hardwareType": "1",
  "sceneNumber": 4,
  "size": "2.13"
}
```

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	
Response Data	data	Json	
Total Records	numberOfElements	Integer	
Total Pages	totalPages	Integer	

Total Records		totalElements	Integer	
Template Information List		content	List	
	Store ID	storeId	Integer	
	Merchant ID	merchantId	Integer	
	Agency ID	agencyId	Integer	
	Template ID	id	Integer	Template Number
	Operator	operateUser	String	
	Screen Type	screenType	String	0:single-sided screen,1:segment screen,2:double-sided screen
	Template Type	sceneNumber	Integer	1:single-producttemplate 2:unbindtemplate3:segment screen template4:multi-producttemplate 6:printtemplate 7:Arrowtemplate
	Template Name	templateName	String	
	Created Time	createdTime	String	
	End time	updateTime	String	

Template Category	attrCategory	String	
Template Attribute	attrName	String	
Number of Products	itemNum	Integer	Number of products that can be bound to the template
Supported ESL Models	modelList	List	
Drawing Layout	drawLayout	Integer	Horizontal or vertical
Width	width	Integer	
Height	height	Integer	
Size	size	String	
Resolution	resolution	String	
Hardware Type (Supported Colors)	hardwareType	Integer	"0": black/white, "1": black/white/red "2": black/white/yellow, "3": multi-color

Response Example (related error code no.: 12 13)

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": {
    "numberOfElements": 1,
    "totalPages": 1,

```

```
"content": [  
  {  
    "id": 3012,  
    "templateNumber": "1620798613384",  
    "screenType": 0,  
    "sceneNumber": 4,  
    "templateName": "1620798529219",  
    "modelId": ",56,",  
    "size": "2.13",  
    "resolution": "250*122",  
    "hardwareType": 1,  
    "hardwareStr": "black/white/red",  
    "drawLayout": 1,  
    "width": 250,  
    "height": 122,  
    "templateVersion": null,  
    "agencyId": 1584608864268,  
    "merchantId": 1584608946854,  
    "merchantName": "shen",  
    "storeId": 1584608946868,  
    "storeName": null,  
    "tempPic": null,  
    "createdTime": "2021-05-12 13:50:13",  
    "flag": 1,  
    "isEnabled": 1,  
    "attrCategory": null,  
    "attrName": null,  
    "specType": 1,  
    "delTime": null,  
    "updateTime": "2021-05-12 13:50:13",  
    "operateUser": "shenbusiness",  
    "itemNum": 3,  
    "isComplete": 1,  
    "missingField": "",  
    "modelList": [  
      "OZKC21S",  
      "ZKC21S-N",  
      "ZKC21V",  
      "ZKC21V-N"  
    ],  
    "tempPicUrls": [  
      "group1/M00/00/E8/rBBYBWCbbJWAWi-cAAAFH-D2JWs639.png"  
    ]  
  },  
  ],  
  "totalElements": 1  
}
```

5.15 Query Merchant Templates

API Definition

URL	http://[server address]/zk/template/listMerchant
Example	http://esl.zkong.com/zk/template/listMerchant
Request Method	POST
Content Type	application/json;charset=utf-8
Token	Add Authorization: [token value returned by login API 3.2] to the request header.
Description	Query ESL templates under a merchant.

Request Parameters

Parameter Name		Parameter Type	Data Type	Required	Notes
Authorization		header	String	true	Authorization token
page		path	Integer	true	Page number (default first page number is 0)
size		path	Integer	true	Page Size
useExternalStoreId		query	Integer	false	useExternalStoreId 0: do not use, 1: use
externalStoreId		query	String (40)	false	External Store ID required when used
Object		body	Object	true	Query condition object
	hardwareType	body	String(1)	false	Supported Colors "0": black/white, "1": black/white/red "2": black/white/yellow, "3": multi-color
	sceneNumber	body	Integer(1)	false	Template Type 1:single-producttemplate 2:unbindtemplate 3:segment screentemplate 4:multi-

					producttemplate 6:printtemplate 7:Arrowtemplate
	size	body	String(1)	false	Template Size

Request Example

Request URL:
<http://es1.zkong.com/zk/template/listMerchant/0/10>
 Request Body:

```
{
  "hardwareType": "1",
  "sceneNumber": 4,
  "size": "2.13"
}
```

Response Parameters

Field Name	Variable Name	Data Type	Notes
Request Successful	success	Boolean	true/false
Status Code	code	Integer	See API error codes for details
Message	message	String	
Response Data	data	Json	
Total Records	numberOfElements	Integer	
Total Pages	totalPages	Integer	
Total Records	totalElements	Integer	
Template Information List	content	List	
Merchant ID	merchantId	Integer	
Agency ID	agencyId	Integer	
Template ID	id	Integer	Template Number

	Operator	operateUser	String	
	Screen Type	screenType	String	0:single-sided screen,1:segment screen,2:double-sided screen
	Template Type	sceneNumber	Integer	1:single-producttemplate 2:unbindtemplate3:segment screentemplate4:multi-producttemplate
	Template Name	templateName	String	
	Created Time	createdTime	String	
	End time	updateTime	String	
	Template Category	attrCategory	String	
	Template Attribute	attrName	String	
	Number of Products	itemNum	Integer	Number of products that can be bound to the template
	Supported ESL Models	modelList	List	
	Drawing Layout	drawLayout	Integer	Horizontal or vertical

	Width	width	Integer	
	Height	height	Integer	
	Size	size	String	
	Resolution	resolution	String	
	Hardware Type (Supported Colors)	hardwareType	Integer	"0": black/white, "1": black/white/red "2": black/white/yell ow, "3": multi- color

Response Example (related error code no.: 12 13)

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": {
    "numberOfElements": 1,
    "totalPages": 1,
    "content": [
      {
        "id": 3012,
        "templateNumber": "1620798613384",
        "screenType": 0,
        "sceneNumber": 4,
        "templateName": "1620798529219",
        "modelId": ",56,",
        "size": "2.13",
        "resolution": "250*122",
        "hardwareType": 1,
        "hardwareStr": "black/white/red",
        "drawLayout": 1,
        "width": 250,
        "height": 122,
        "templateVersion": null,
        "agencyId": 1584608864268,
        "merchantId": 1584608946854,

```

```

"merchantName": "shen",
"storeId": 1584608946868,
"storeName": null,
"tempPic": null,
"createTime": "2021-05-12 13:50:13",
"flag": 1,
"isEnable": 1,
"attrCategory": null,
"attrName": null,
"specType": 1,
"delTime": null,
"updateTime": "2021-05-12 13:50:13",
"operateUser": "shenbusiness",
"itemNum": 3,
"isComplete": 1,
"missingField": "",
"modelList": [
  "OZKC21S",
  "ZKC21S-N",
  "ZKC21V",
  "ZKC21V-N"
],
"tempPicUrls": [
  "group1/M00/00/E8/rBBYBWCbbJWAWi-cAAAFH-D2JWs639.png"
]
}
],
"totalElements": 1
}
}

```

6. Organization and Permission APIs

6.1 Add Store

API Definition

URL	http://[server address]/zk/store/add
Example	http://esl.zkong.com/zk/store/add
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Add a store. When merchant data synchronization is enabled, store ESLs can bind merchant products. If store product attributes are empty, non-empty corresponding merchant attributes are automatically used.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
storeName	body	String(100)	true	Store Name
dataCombination	body	Integer(1)	true	Whether to enable merchant data synchronization (0: no, 1: yes)
externalStoreId	body	String(40)	false	External Store ID
contacts	body	String(40)	false	Contact Person
phone	body	String(40)	false	Phone
mailbox	body	String(60)	false	Email
address	body	String(200)	false	Address
comment	body	String(200)	false	Notes
label	body	String(60)	false	Label
orgId	body	Integer(11)	false	Organization ID
onlineTime	body	String(60)	false	Example: 2025-02-26 15:35:19

Request Example

Request URL:

<http://es1.zkong.com/zk/store/add>

Request Body:

```
{
  "storeName": "Hangzhou Store 1",
  "dataCombination": 0,
  "externalStoreId": "123159",
  "contacts": "",
  "phone": "17602549811",
  "mailbox": "123159@163.com",
  "address": "No. 546 Renmin Road, Xihu District, Hangzhou",
  "comment": "Remarks",
  "label": "Little Swan",
}
```

```
"onlineTime": "2025-02-26 15:35:19",  
"orgId": 1  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 2 4 41 90 91 92)

```
{  
  "success":true,  
  "code":14010,  
  "message":"Store added successfully",  
  "data":{  
    "id":0,  
    "storeId":1632742186074,  
    "parentId":"0076",  
    "storeName":"123159",  
    "externalStoreId":"123159",  
    "status":1,  
    "agencyId":1555414082383,  
    "merchantId":1615365123378,  
    "merchantName":"hsb",  
    "supervisePhone":null,  
    "comment":"123159",  
    "useMerchantTemp":null,  
    "templateVersion":null,  
    "createdTime":"2021-09-27 19:29:46",  
    "address":"123159",  
    "mailbox":"123159",  
    "phone":"123159",  
  }  
}
```

```

    "contacts": "123159",
    "flag": 1
  }
}

```

Example 2

```

{
  "success": false,
  "code": 13030,
  "message": "You do not have permission to access this",
  "data": null,
  "translate": true,
  "originalMessage": null
}

```

6.2 Update Store Information

API Definition

URL	http://[server address]/zk/store/update/{id}
Example	http://esl.zkong.com/zk/store/update/1546
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Update a store. Only the provided attributes are updated.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer(11)	true	Store ID
storeName	body	String(100)	true	Store Name
externalStoreId	body	String(40)	false	External Store ID
contacts	body	String(40)	false	Contact Person
phone	body	String(40)	false	Phone

mailbox	body	String(60)	false	Email
address	body	String(100)	false	Address
comment	body	String(200)	false	Notes
label	body	String(60)	false	Label
orgId	body	Integer(11)	false	Organization ID
onlineTime	body	String(60)	false	Example: 2025-02-26 15:35:19

Request Example

Request URL:
<http://esl.zkong.com/zk/store/update/1546>

Request Body:

```
{
  "storeName": "Hangzhou Store 1",
  "externalStoreId": "123159",
  "contacts": "",
  "phone": "17602549811",
  "mailbox": "123159@163.com",
  "address": "No. 546 Renmin Road, Xihu District, Hangzhou",
  "comment": "Remarks",
  "label": "Little Swan",
  "onlineTime": "2025-02-26 15:35:19",
  "orgId": 1
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 3 13 90 93)

```
{
  "success": true,
  "code": 14012,
  "message": "Store updated successfully",
  "data": {
    "id": 17043,
    "storeId": 1632742186074,
    "parentId": "0076",
    "storeName": "123",
    "externalStoreId": "123159",
    "status": 1,
    "agencyId": 1555414082383,
    "merchantId": 1615365123378,
    "merchantName": "abc",
    "supervisePhone": null,
    "comment": "123159",
    "useMerchantTemp": null,
    "templateVersion": null,
    "createdTime": "2021-09-27 19:29:46",
    "address": "123159",
    "mailbox": "123159",
    "phone": "1",
    "contacts": "1",
    "flag": 1
  },
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 14054,
  "message": "The external store ID already exists under this merchant",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

6.3 Query Store List

API Definition

URL	http://[server address]/zk/store/storeList
Example	http://esl.zkong.com/zk/store/storeList
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query the list of stores that the current user has permission to access.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token

Request Example

Request URL:
http://esl.zkong.com/zk/store/storeList
 Request Body:
None

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 3)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": [
    {
      "id": 797,
      "storeId": 1607566698661,
      "parentId": "0001",

```

```
"storeName": "Store 1",
"externalStoreId": "1001",
"status": 1,
"agencyId": 2147483649,
"merchantId": 1607566698642,
"merchantName": "hu9611",
"comment": "",
"createTime": "2020-12-10 10:18:18",
"address": "",
"mailbox": "",
"phone": "18721542154",
"contacts": "Li Qiang",
"onlineTime": "2025-02-26 15:35:19",
"flag": 1
},
{
  "id": 17021,
  "storeId": 1631504419546,
  "parentId": "0002",
  "storeName": "Store 2",
  "externalStoreId": "1002",
  "status": 1,
  "agencyId": 2147483649,
  "merchantId": 1607566698642,
  "merchantName": "hu9611",
  "comment": "",
  "createTime": "2021-09-13 11:40:19",
  "address": "",
  "mailbox": "455126456@qq.com",
  "phone": "176025471254",
  "contacts": "Li Hua",
  "onlineTime": "2025-02-26 15:35:19",
  "flag": 1
}
],
"translate": true,
"originalMessage": null
}
```

6.4 Query Role List

API Definition

URL	http://[server address]/zk/role/queryRoleList
Example	http://esl.zkong.com/zk/role/queryRoleList
Request Method	POST
Content Type	application/json;charset=utf-8

Description	Query the list of roles that the current user has permission to access.
--------------------	---

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
merchantId	body	Long	true	Merchant ID

Request Example

Request URL:
<http://es1.zkong.com/zk/role/queryRoleList>
Request Body:

```
{
  "merchantId": 1629968747662
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example

```
{
  "success": true,
  "code": 10029,
  "message": "Role list queried successfully",
  "data": [
    {
      "id": 3,
      "name": "Merchant Super Administrator",
      "status": 0,
      "isAble": 0
    }
  ],
}
```

```
{
  "id": 743,
  "name": "Merchant Administrator",
  "status": 1,
  "isAble": 1
},
{
  "id": 1329,
  "name": "Store Manager",
  "status": 1,
  "isAble": 1
},
{
  "id": 1474,
  "name": "Operator",
  "status": 1,
  "isAble": 1
}
],
"translate": true,
"originalMessage": null
}
```

6.5 Add Employee Account

API Definition

URL	http://[server address]/zk/user/add
Example	http://esl.zkong.com/zk/user/add
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Add Employee Account

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
account	body	String	true	Employee Account (Note: supports 6-20 digits, letters, and '_')

				@.'; case-insensitive)
password	body	String	true	Password (Note: at least 8 characters, including at least 3 of the following: numbers, uppercase letters, lowercase letters, and special characters)
name	body	String	true	Employee Name
phone	body	String	false	Mobile number (supports 0-20 digits)
roleId	body	Integer	true	Role ID
storeIds	body	List<Long>	true	Data Permissions (store ID list)
allStorePermission	body	Integer	true	Automatically add permissions for newly created stores (0: no, 1: yes)

Request Example

Request URL:
<http://es1.zkong.com/zk/user/add>

Request Body:

```
{
  "account": "example123",
  "name": "example",
  "password": "zkong123+",
  "phone": "",
  "roleId": 743,
  "storeIds": [
    1683872631979,
    1630484444737
  ],
  "allStorePermission": 1
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false

code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example

```
{
  "success": true,
  "code": 14019,
  "message": "User added successfully",
  "data": {
    "id": 1056,
    "account": "231231",
    "password": "ae2ceda262f46381624a70c1e5683f9b",
    "roleId": 743,
    "lastLoginTime": "2024-03-20 13:48:01",
    "agencyId": 1629967488587,
    "agencyName": " zzhd11",
    "merchantId": 1629968747662,
    "merchantName": "hzx1234",
    "enable": 1,
    "name": "123213",
    "phone": "",
    "email": null,
    "accountNum": "0013",
    "loginErrorNum": 0,
    "lastErrorTime": "1710913681000",
    "allStorePermission": 0,
    "createTime": "2024-03-20 13:48:01",
    "deleted": 0,
    "roleName": null,
    "token": null,
    "storeIdList": null,
    "agencyIdList": null,
    "passwordUpdateTime": 1710913681000
  },
  "translate": true,
  "originalMessage": null
}
```

6.6 Query Employee List by Page

API Definition

URL	http://[server address]/zk/user/findUserList
Example	http://esl.zkong.com/zk/user/findUserList
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query the list of employees within the current user's permissions.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
account	body	String	false	Employee Account
name	body	String	false	Employee Name
currentPage	body	Integer	true	Current Page, starting from 1
pageSize	body	Integer	true	Page Size
roleIds	body	List<Integer>	false	Role ID List
enables	body	List<Integer>	false	Status list (empty: all 0: disable 1: enable)

Request Example

```

Request URL:
http://esl.zkong.com/zk/user/findUserList
Request Body:
{
  "name": "",
  "currentPage": 1,
  "pageSize": 10,
  "roleIds": [],
  "enables": [
    1
  ],
  "account": null
}

```

Response Parameters

Parameter Name	Description	Data Type	Notes
----------------	-------------	-----------	-------

success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example

```
{
  "success": true,
  "code": 14016,
  "message": "User list queried by page successfully",
  "data": {
    "total": 9,
    "userVos": [
      {
        "user": {
          "id": 57,
          "account": "hzx2587",
          "password": null,
          "roleId": 743,
          "createTime": "2021-08-31 16:34:51",
          "lastLoginTime": "2023-10-30 11:22:10",
          "token": null,
          "merchantId": null,
          "merchantName": null,
          "storeId": null,
          "storeName": null,
          "enable": 0,
          "flag": null,
          "name": "hzx25872",
          "phone": "",
          "email": null,
          "accountNum": "0002",
          "agencyId": null,
          "agencyName": null,
          "loginErrorNum": null,
          "lastErrorTime": null,
          "allStorePermission": null,
          "passwordUpdateTime": null
        },
        "roleName": "Merchant Administrator",
        "agencyDtoVos": null,
        "isZong": null
      }
    ]
  }
}
```

```
},
{
  "user": {
    "id": 50,
    "account": "hzx123",
    "password": null,
    "roleId": 3,
    "createTime": "2021-08-26 17:05:47",
    "lastLoginTime": "2024-03-20 11:06:31",
    "token": null,
    "merchantId": null,
    "merchantName": null,
    "storeId": null,
    "storeName": null,
    "enable": 1,
    "flag": null,
    "name": "hzx8523",
    "phone": "",
    "email": null,
    "accountNum": "0001",
    "agencyId": null,
    "agencyName": null,
    "loginErrorNum": null,
    "lastErrorTime": null,
    "allStorePermission": null,
    "passwordUpdateTime": null
  },
  "roleName": "Merchant Super Administrator",
  "agencyDtoVos": null,
  "isZong": null
}
],
},
"translate": true,
"originalMessage": null
}
```

6.7 Edit Employee Account Information

API Definition

URL	http://[server address]/zk/user/updateUser
Example	http://esl.zkong.com/zk/user/updateUser
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Update employee account.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	body	Integer	true	Account ID to modify
account	body	String	true	Username
password	body	String	false	Password (Note: at least 8 characters, including at least 3 of the following: numbers, uppercase letters, lowercase letters, and special characters)
name	body	String	false	Employee Name
phone	body	String	false	Mobile number (supports 0-20 digits)
roleId	body	Integer	true	Role ID
storeIds	body	List<Long>	true	Data Permissions (store ID list)
allStorePermission	body	Integer	true	Automatically add permissions for newly created stores (0: no, 1: yes)

Request Example

Request URL:
<http://es1.zkong.com/zk/user/add>

Request Body:

```
{
  "id": 256,
  "name": "example",
  "password": "zkong123+",
  "phone": "",
  "roleId": 743,
  "storeIds": [
```

```
1683872631979,  
1630484444737  
],  
"allStorePermission": 1  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example

```
{  
  "success": true,  
  "code": 10000,  
  "message": "Operation successful",  
  "data": {  
    "success": true,  
    "code": 10000,  
    "message": "success",  
    "data": {  
      "id": 1056,  
      "account": "231231",  
      "password": "ae2ceda262f46381624a70c1e5683f9b",  
      "roleId": 743,  
      "lastLoginTime": "2024-03-20 13:48:01",  
      "agencyId": 1629967488587,  
      "agencyName": " zzhd11",  
      "merchantId": 1629968747662,  
      "merchantName": "hzx1234",  
      "enable": 1,  
      "name": "123213",  
      "phone": "123123",  
      "email": null,  
      "accountNum": "0013",  
      "loginErrorNum": 0,  
      "lastErrorTime": "1710916276000",  
      "allStorePermission": 0,  
      "createTime": "2024-03-20 13:48:01",  
      "deleted": 0,  
      "roleName": null,  
    }  
  }  
}
```

```
"token": null,  
"storeIdList": null,  
"agencyIdList": null,  
"passwordUpdateTime": 1710916276000  
}  
},  
"translate": true,  
"originalMessage": null  
}
```

6.8 Disable Employee Account

API Definition

URL	http://[server address]/zk/user/disable/{id}
Example	http://esl.zkong.com/zk/user/disable/1024
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Disable an employee account. After it is disabled, the account cannot log in.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Account ID

Request Example

Request URL:
http://esl.zkong.com/zk/user/disable/1024
Request Body:
None

Response Example

```
{  
  "success": true,  
  "code": 10000,  
  "message": "Operation successful",  
  "data": null,  
  "translate": true,  
  "originalMessage": "success"  
}
```

```
}

```

6.9 Enable employee account.

API Definition

URL	http://[server address]/zk/user/enable/{id}
Example	http://esl.zkong.com/zk/user/enable/1024
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Enable employee account.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Account ID

Request Example

Request URL:
<http://esl.zkong.com/zk/user/enable/1024>
 Request Body:
 None

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

6.10 Delete Employee Account

API Definition

URL	http://[server address]/zk/user/delete/{id}
Example	http://esl.zkong.com/zk/user/delete/1024
Request Method	DELETE
Content Type	application/json;charset=utf-8
Description	Delete employee account. This action cannot be recovered.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Account ID

Request Example

Request URL:
http://esl.zkong.com/zk/user/delete/1024
Request Body:
None

Response Example

```
{
  "success": true,
  "code": 14017,
  "message": "User deleted successfully",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

6.11 Query Organization Information List

API Definition

URL	http://[server address]/zk/organization/getList
Example	http://esl.zkong.com/zk/organization/getList

Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query the organization information list for the merchant to which the current user belongs.

Request Parameters

None

Request Example

Request URL:
<http://esl.zkong.com/zk/organization/getList>
 Request Body:
 None

Organization Information Field Definitions

id	Organization Information ID
name	Organization Name
merchantId	Merchant ID
parentId	Parent Node ID
level	Level
path	Path

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": [
    {
      "id": 1,
      "name": "East China Region",
      "merchantId": 1676959167307,
      "parentId": 0,
      "level": 1,
      "path": "1",
      "createTime": "2025-02-19 11:27:26",
      "updateTime": "2025-02-19 11:27:27",
      "createUserId": 294,
      "updateUserId": null
    }
  ],
}
```

```
{
  "id": 2,
  "name": "North China Region",
  "merchantId": 1676959167307,
  "parentId": 0,
  "level": 1,
  "path": "2",
  "createTime": "2025-02-19 11:27:46",
  "updateTime": "2025-02-19 11:27:46",
  "createUserId": 294,
  "updateUserId": null
},
{
  "id": 3,
  "name": "Southeast China Region",
  "merchantId": 1676959167307,
  "parentId": 1,
  "level": 2,
  "path": "1/3",
  "createTime": "2025-02-19 11:27:56",
  "updateTime": "2025-02-19 11:31:26",
  "createUserId": 294,
  "updateUserId": 294
}
],
"translate": true,
"originalMessage": null
}
```

7. LCD Operations and Information Query

7.1 Bind Product to LCD Device

API Definition

URL	http://[server address]/lcd/layout/bindItem
Example	http://esl.zkong.com/lcd/layout/bindItem
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Bind products to an LCD device for content playback. Up to 10 products can be bound.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
duid	body	String(64)	true	Device DUID
bindList	body	List<Object>(10)	true	Bound Product List
barcode	body	String(80)	true	Product Code
sequence	body	Integer(2)	true	Product sequence number (must increase sequentially according to the number of bound products, starting from 1)

Request Example

Request URL:

<http://es1.zkong.com/lcd/layout/bindItem>

Request Body:

```
{
  "duid": "EDCD778FA7D3",
  "bindList": [
    {
      "barcode": "6943941100337",
      "sequence": 1
    },
    {
      "barcode": "6943941100352",
      "sequence": 2
    },
    {
      "barcode": "6943941100368",
      "sequence": 3
    },
    {
      "barcode": "6943941100347",
      "sequence": 4
    },
    {
      "barcode": "694411005267",
```

```
"sequence": 5
}
]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 41 94 95)

```
{
  "success": true,
  "code": 10000,
  "message": "success",
  "data": null
}
```

Example 2

```
{
  "success": false,
  "code": 52023003,
  "message": "LCD device does not exist",
  "data": null
}
```

7.2 Query LCD Binding Information

API Definition

URL	http://[server address]/lcd/layout/bindList
Example	http://esl.zkong.com/lcd/layout/bindList
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query product information and device information bound to the LCD.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
duid	body	String(64)	true	Device DUID

Request Example

```
Request URL:  
http://esl.zkong.com/lcd/layout/bindList  
Request Body:  
{  
  "duid": "3C866C9ABD92"  
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message

data	Response Data	Json	
------	---------------	------	--

Response Example (related error code no.: 94 95)

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": {
    "bindList": [
      {
        "sequence": 1,
        "barcode": "1213456",
        "itemTitle": null
      },
      {
        "sequence": 2,
        "barcode": "1231534",
        "itemTitle": null
      },
      {
        "sequence": 3,
        "barcode": "125132123",
        "itemTitle": null
      },
      {
        "sequence": 4,
        "barcode": "132",
        "itemTitle": null
      },
      {
        "sequence": 5,
        "barcode": "130",
        "itemTitle": null
      }
    ],
    "device": {
      "id": 202211,
      "name": "hzxtest",
      "duid": "3C866C9ABD92",
      "agencyId": 1629967488587,
      "merchantId": 1629968747662,
      "storeId": 1683872631979,
      "code": null,
      "modelId": 8,
      "brightness": 100,
      "volume": 50,
      "space": "0.096/0.094",
      "comment": ""
    }
  }
}
```

```
"lastCommuTime": 1701240874000,
"lastReportTime": 1701240874000,
"status": 0,
"appVersion": "0.16.013",
"systemVersion": "Flythings OS 2021/11/24(6701408)",
"switchConfig": null,
"switchConfigStatus": null,
"createTime": "2023-11-27 13:56:46",
"updateTime": "2023-11-29 14:58:56",
"createUserId": 50,
"updateUserId": 50,
"qrCodeStatus": 0,
"codeType": 1,
"barcode": "",
"uuid": "01D:21:27:15",
"networkState": 0,
"upgradeStatus": 5,
"ssid": "AP_2.4G_A310",
"macAddress": "A0A3B960A310",
"currPublicationId": "2711",
"ip": "192.168.11.149",
"rotationAngle": 90,
"runTimeSec": 77250
},
"thumbnailUrl": null
}
}
```

Example 2

```
{
  "success": false,
  "code": 52023003,
  "message": "LCD device does not exist",
  "data": null
}
```

7.3 Issue LCD Device Command

API Definition

URL	http://[server address]/lcd/device/sendCmd
Example	http://esl.zkong.com/lcd/device/sendCmd
Request Method	POST
Content Type	application/json;charset=utf-8

Description	Issue operation commands to LCD devices.
--------------------	--

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
duids	body	List<String>	true	Device DUID List
type	body	Integer	true	Command type. See Appendix 3 for details
value	body	String	false	Command value. Required for some commands

Request Example

Request URL:
<http://es1.zkong.com/lcd/device/sendCmd>

Request Body:

```
{
  "duids": [
    "3C866C9ABD92"
  ],
  "type": 6,
  "value": "80"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": "success",
  "data": null
}
```

7.4 Query LCD Device List

API Definition

URL	http://[server address]/lcd/device/getList
Example	http://esl.zkong.com/lcd/device/getList
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query the LCD device list by condition.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
page	query	Integer	false	Default: 1
size	query	Integer	false	Default: 10
deviceType	body	Integer(1)	false	Device Type1: commercial displaydevice 2: LCDdevice
keyword	body	String	false	Search keyword. Supports DUID, device name, product code, and device barcode.
status	body	Integer(1)	false	Online status 0: offline, 1: online
storeIds	body	List<Long>	false	Store list (if not provided, all merchant data is queried)

Request Example

```
Request URL:
http://esl.zkong.com/lcd/device/getList
Request Body:
{
  "deviceType": 2,
  "keyword": "",
  "status": 1,
  "storeIds": [
    1683872631979
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 1)

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": {
    "totalElements": 1, //Total Records
    "list": [
      {
        "id": 202211, //deviceId
        "labelList": null, //Device Labels
        "name": "hzxtest", //Device Name
        "duid": "3C866C9ABD92", //deviceDuid
        "agencyId": 1629967488587,
        "merchantId": 1629968747662,
        "storeId": 1683872631979,
        "code": null, //Device Code
        "modelId": 8, //modelID
        "brightness": 100, //Brightness
      }
    ]
  }
}
```

```

"volume": 50, //Volume
"space": "0.096/0.094", //Memory Space
"comment": "",
"lastCommuTime": "2023-11-29 16:54:34", //last communication time
"lastReportTime": "2023-11-29 16:54:34", //last report time
"status": 0, //Online Status
"appVersion": "0.16.013", //app version
"systemVersion": "Flythings OS 2021/11/24(6701408)", //system version
"createTime": "2023-11-27 15:56:46",
"updateTime": "2023-11-29 16:58:56",
"createUserId": 50,
"updateUserId": 50,
"publishNum": 0,
"model": "ZKS101", //model
"oemModel": "ZKS101", //oemmodel
"resolution": "800*1280", //Resolution
"publicationNum": null,
"size": "10.1",
"previewInfo": null,
"lastApkVersion": "1.00.002",
"lastSystemVersion": "--",
"upgradeStatus": 5,
"qrCodeStatus": 0,
"codeStatusConfig": null,
"storeName": "123456",
"codeType": 1,
"codeTypeConfig": null,
"barcode": "",
"uuid": "01D:21:27:15",
"networkState": 0, //Network Mode
"deviceType": 2,
"merchantName": null,
"ip": "192.168.11.149",
"bindInfo": "null-1213456、null-1231534、null-125132123、null-132、null-130",
"rotationAngle": 90,
"runTimeSec": 77250
}
]
}
}

```

7.5 Upload merchant image and audio materials.

API Definition

URL	http://[server address]/lcd/content/upload
------------	--

Example	http://esl.zkong.com/lcd/content/upload
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Upload merchant image and audio materials.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
file	body	MultipartFile	true	Image files ("jpeg", "bmp", "gif", "png", "jpg") or audio files (mp3, wma, wav), maximum 10 MB.
description	query	String	false	Content Description

Request Example

Request URL:
<http://esl.zkong.com/lcd/content/upload>
 Request Body:

```
{
  "file": "<uploadfile>"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	Relative File Path

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": "group1/M00/05/7E/wKgAa2XypxmAEtMMAAHR6uZPISc924.png"
}
```

7.6 Upload merchant video materials.

API Definition

URL	http://[server address]/lcd/content/uploadVideo
Example	http://esl.zkong.com/lcd/content/uploadVideo
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Upload merchant video materials.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
file	body	MultipartFile	true	Video files, maximum 100 MB. Supported formats: mp4, avi, rmvb, flv, and 3gp.
description	query	String	false	Content Description

Request Example

Request URL:
http://esl.zkong.com/lcd/content/uploadVideo?description=Description

Request Body:

```
{
  "file": "<uploadfile>"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	Relative File Path

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": "group1/M00/05/7E/wKgAa2XypxmAEtMKNGHJr6uZPISc125.mp4"
}
```

7.7 Add LCD Device to Store

API Definition

URL	http://[server address]/lcd/device/save
Example	http://esl.zkong.com/lcd/device/save
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Add the device to the specified store.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
duid	body	String	true	Device DUID
storeId	body	Long	true	Store ID

modelId	body	Integer	true	Device Model
name	body	String	true	Device Name
barcode	body	String	false	Device Barcode
comment	body	String	false	Notes

Request Example

Request URL:
<http://esl.zkong.com/lcd/device/save>

Request Body:

```
{
  "duid": "SZKS101241203000434",
  "name": "SZKS101241203000434",
  "storeId": "1683872631979",
  "modelId": 8,
  "barcode": "",
  "comment": ""
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	Relative File Path

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": "success"
}
```

7.8 Issue Images and Videos to Specified LCD Device

API Definition

URL	http://[server address]/lcd/erp/publication/add
Example	http://esl.zkong.com/lcd/erp/publication/add
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Issue images and videos to a specified LCD device for playback. Scheduled playback, resource layout, and carousel playback of multiple resources in the same area are supported.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer(1)	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
duid	body	String	true	Unique Device Identifier
storeId	body	Long	true	Store ID (required when external store ID is not used)
playTime	body	Long	false	Scheduled playback time (second-level timestamp). If not provided, playback starts immediately.
playContents	body	List<Content>	true	Playback Content
abscissa	body	Integer	true	X-coordinate, image/video start position: distance from the left edge
ordinate	body	Integer	true	Y-coordinate, image/video start position: distance from the top edge
width	body	Integer	true	Image/Video Width
height	body	Integer	true	Image/Video Height
url	body	String	true	Image/video resource URL (multiple values can be separated by commas)

	type	body	Integer	true	Resource Type 1 video (supports mp4, avi, rmvb, flv, 3gp) 2 image (supports jpeg, bmp, gif, png, jpg)
	refreshTime	body	Integer	No	Image carousel interval, default: 30s (effective when multiple image resources are used)

Request Example

Request URL:

<http://esl.zkong.com/lcd/erp/publication/add>

Request Body:

```
{
  "duid": "SZKS101241023000538",
  "storeId": 1683872631979,
  "playTime": 1748312100,
  "playContents": [
    {
      "abscissa": 0,
      "ordinate": 0,
      "width": "800",
      "height": "640",
      "url": "https://esl-hz.zkong.com/group1/M00/A9/21/wKgAIGgq-Q-AEcKYABV8vGVNnB8444.png,https://esl-hz.zkong.com/group1/M00/A9/21/wKgAIGgq-QaAPEyuABjv-nujeAk103.png",
      "type": "2",
      "refreshTime": 5
    },
    {
      "abscissa": 0,
      "ordinate": 640,
      "width": "800",
      "height": "640",
      "url": "https://esl-hz.zkong.com/group1/M00/A9/21/wKgAIGgq-QaAPEyuABjv-nujeAk103.png",
      "type": "2"
    }
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false

code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	String	Relative File Path

Response Example

```
{
  "success": true,
  "code": 10000,
  "message": null,
  "data": "success"
}
```

8. Promotion Strategy APIs

8.1 Create Promotion Strategy

API Definition

URL	http://[server address]/zk/strategy/create
Example	http://esl.zkong.com/zk/strategy/create
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Create a promotion strategy.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
useExternalStoreId	query	Integer(1)	false	useExternalStoreId 0: do not use, 1: use

externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
name	body	String(50)	true	Strategy Name
startDate	body	Long	true	Validity Start Timestamp
endDate	body	Long	true	Validity End Timestamp
templateAttrCategory	body	String(50)	true	Promotion Template Category
templateAttr	body	String(50)	true	Promotion Template Attribute
triggerType	body	Integer	true	Trigger Method 1: fixed cycle, 2: always triggered
periodType	body	Integer	true	Trigger Period Type 0: daily, 1: weekly, 2: monthly
periodValue	body	int[]	true	Trigger Period Value When periodType = 0, pass an empty array When periodType = 1, available values are 1- 7. Use 1 for Sunday, 2 for Monday, and so on. When periodType = 2, available values are 1- 31, representing the day of the month
periodTimes	body	String[]	true	Trigger time period, i.e. start time and end time, in HH:mm:ss format
selectFieldNameNum	body	Int[]	true	Optional field array to

					set besides promotion sales price and promotion member price. Select up to 5 fields. Pass an empty array if none are selected. Available values: 0-19 (0: Sales Unit 1: Product Grade 2: Place of Origin 3: Original Price 4: Promotion Text 5: extended1 and so on, 19: extended field 15)
	itemActions	body	List	true	Execution Object List
	itemId	body	Long	true	Product ID
	fieldValues	body	Map	true	Promotion product attribute value set; at least one promotion attribute value is required
	price	body	String	false	Promotion Sales Price
	memberPrice	body	String	false	Promotion Member Price
	unit	body	String	false	Sales Unit
	classLevel	body	String	false	Product Grade
	productArea	body	String	false	Place of Origin
	originalPrice	body	String	false	Original Price
	promotionText	body	String	false	Promotion Text
	custFeature1	body	String	false	Extended Field 1

	custFeature2	body	String	false	Extended Field 2
	custFeature3	body	String	false	Extended Field 3
	custFeature4	body	String	false	Extended Field 4
	custFeature5	body	String	false	Extended Field 5
	custFeature6	body	String	false	Extended Field 6
	custFeature7	body	String	false	Extended Field 7
	custFeature8	body	String	false	Extended Field 8
	custFeature9	body	String	false	Extended Field 9
	custFeature10	body	String	false	Extended Field 10
	custFeature11	body	String	false	Extended Field 11
	custFeature12	body	String	false	Extended Field 12
	custFeature13	body	String	false	Extended Field 13
	custFeature14	body	String	false	Extended Field 14
	custFeature15	body	String	false	Extended Field 15

Request Example

Request URL:

<http://es1.zkong.com/zk/strategy/create?useExternalStoreId=1&externalStoreId=1684845118494>

Request Body:

```
{
  "name": "Limited-time Discount Strategy",
  "startDate": 1712678400000,
  "endDate": 1714492799000,
  "templateAttr": "default",
  "templateAttrCategory": "default",
  "triggerType": 1,
  "periodType": 1,
  "periodValue": [
    2,
    7,
    1
  ],
  "selectFieldNameNum": [5, 4],
  "itemActions": [
```

```
{
  "itemId": 11184969707,
  "fieldValues": {
    "price": "3.5",
    "memberPrice": "2.5",
    "unit": null,
    "classLevel": null,
    "productArea": null,
    "originalPrice": "4.35",
    "promotionText": "Limited-time Special Offer",
    "custFeature1": "Until the end of the month",
    "custFeature2": null,
    "custFeature3": null,
    "custFeature4": null,
    "custFeature5": null,
    "custFeature6": null,
    "custFeature7": null,
    "custFeature8": null,
    "custFeature9": null,
    "custFeature10": null,
    "custFeature11": null,
    "custFeature12": null,
    "custFeature13": null,
    "custFeature14": null,
    "custFeature15": null
  }
},
  "periodTimes": [
    "19:00:00",
    "21:00:00"
  ]
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Integer	Promotion Strategy ID

Response Example (related error code no.: 12 13 41 45 96 97 98 101)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": 457,
  "translate": true,
  "originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 21013,
  "message": "The current strategy name already exists",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

8.2 Edit Promotion Strategy

API Definition

URL	http://[server address]/zk/strategy/update
Example	http://esl.zkong.com/zk/strategy/update
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Edit a promotion strategy.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	body	Integer	true	Strategy ID
merchantId	body	Long	true	Merchant ID

storeId	body	Long	true	Store ID
Name	body	String(50)	true	Strategy Name
startDate	body	Long	true	Validity Start Timestamp
endDate	body	Long	true	Validity End Timestamp
templateAttrCategory	body	String(50)	true	Promotion Template Category
templateAttr	body	String(50)	true	Promotion Template Attribute
triggerType	body	Integer	true	Trigger Method 1: fixed cycle, 2: always triggered
periodType	body	Integer	true	Trigger Period Type 0: daily, 1: weekly, 2: monthly
periodValue	body	int[]	true	Trigger Period Value When periodType = 0, pass an empty array When periodType = 1, available values are 1-7. Use 1 for Sunday, 2 for Monday, and so on. When periodType = 2, available values are 1-31, representing the day of the month
periodTimes	body	String[]	true	Trigger time period, i.e. start time and end time, in HH:mm:ss format
selectFieldNameNum	body	Int[]	true	Optional field array to set besides promotion

					sales price and promotion member price. Select up to 5 fields. Pass an empty array if none are selected. Available values: 0-19 (0: Sales Unit 1: Product Grade 2: Place of Origin 3: Original Price 4: Promotion Text 5: extended1 and so on, 19: extended field 15)
	itemActions	body	List	true	Execution Object List
	itemId	body	Long	true	Product ID
	fieldValues	body	Map	true	Promotion product attribute value set; at least one promotion attribute value is required
	price	body	String	false	Promotion Sales Price
	memberPrice	body	String	false	Promotion Member Price
	unit	body	String	false	Sales Unit
	classLevel	body	String	false	Product Grade
	productArea	body	String	false	Place of Origin
	originalPrice	body	String	false	Original Price
	promotionText	body	String	false	Promotion Text
	custFeature1	body	String	false	Extended Field 1
	custFeature2	body	String	false	Extended Field 2

	custFeature3	body	String	false	Extended Field 3
	custFeature4	body	String	false	Extended Field 4
	custFeature5	body	String	false	Extended Field 5
	custFeature6	body	String	false	Extended Field 6
	custFeature7	body	String	false	Extended Field 7
	custFeature8	body	String	false	Extended Field 8
	custFeature9	body	String	false	Extended Field 9
	custFeature10	body	String	false	Extended Field 10
	custFeature11	body	String	false	Extended Field 11
	custFeature12	body	String	false	Extended Field 12
	custFeature13	body	String	false	Extended Field 13
	custFeature14	body	String	false	Extended Field 14
	custFeature15	body	String	false	Extended Field 15

Request Example

Request URL:
<http://es1.zkong.com/zk/strategy/update>

Request Body:

```
{
  "id": 457,
  "name": "Limited-time Discount Strategy - New",
  "startDate": 1712678400000,
  "endDate": 1714492799000,
  "templateAttr": "default",
  "templateAttrCategory": "default",
  "triggerType": 1,
  "periodType": 1,
  "periodValue": [
    2,
    6,
    1
  ],
  "selectFieldNameNum": [5, 4],
  "itemActions": [
    {
      "itemId": 11184969707,
```

```

"fieldValues": {
  "price": "3.5",
  "memberPrice": "2.5",
  "unit": null,
  "classLevel": "level",
  "productArea": "China",
  "originalPrice": "4.35",
  "promotionText": "Limited-time Special Offer",
  "custFeature1": "Until the end of the month",
  "custFeature2": null,
  "custFeature3": null,
  "custFeature4": null,
  "custFeature5": null,
  "custFeature6": null,
  "custFeature7": null,
  "custFeature8": null,
  "custFeature9": null,
  "custFeature10": null,
  "custFeature11": null,
  "custFeature12": null,
  "custFeature13": null,
  "custFeature14": null,
  "custFeature15": null
}
],
"periodTimes": [
  "19:00:00",
  "21:00:00"
]
}

```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data		null

Response Example (related error code no.: 41 96 97 98 100 101 102)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 21013,
  "message": "The current strategy name already exists",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

8.2.1 Edit Promotion Strategy

API Definition

URL	http://[server address]/zk/strategy/updateStrategyIncrement
Example	http://esl.zkong.com/zk/strategy/updateStrategyIncrement
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Use the execution list data to incrementally add strategy products without overwriting existing data.

Request parameters are the same as 8.2

Request example is the same as 8.2

Response parameters are the same as 8.2

Response Example (related error code no.: 41 96 97 98 100 101 102)

Example 2

8.3 Query Store Strategies by Page

API Definition

URL	http://[server address]/zk/strategy/list/{page}/{size}
Example	http://esl.zkong.com/zk/strategy/list/1/10?isValid=true
Request Method	POST
Content Type	application/json;charset=utf-8
Description	Query store strategies by page. Supports querying current strategies and expired strategies.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
page	path	Integer	true	Page Number, starting from 1
size	path	Integer	true	Page Size
isValid	query	boolean	true	true: current strategy false: expired strategy
useExternalStoreId	query	Integer(1)	false	useExternalStoreId 0: do not use, 1: use
externalStoreId	query	String(40)	false	External Store ID required when used
storeId	body	Integer(20)	true	Store ID
name	body	String(50)	false	Strategy name search keyword

Request Example

Request URL:

http://esl.zkong.com/zk/strategy/list/1/10?isValid=true

Request Body:

```
{
  "storeId": 1684845118494,
```

```
"name": "Limited Time"
}
```

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 12 13 45)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "list": [
      {
        "id": 457,
        "name": "Limited-time Discount Strategy - New",
        "agencyId": 1684844103168,
        "merchantId": 1684844150922,
        "storeId": 1684845118494,
        "code": "2240457",
        "templateAttrCategory": "default",
        "templateAttr": "default",
        "startDate": "2024-04-10T00:00:00.000+08:00",
        "endDate": "2024-04-30T23:59:59.000+08:00",
        "status": 1,
        "creator": "yintai",
        "createdTime": "2024-04-11T14:30:12.000+08:00",
        "updateTime": "2024-04-11T16:54:19.000+08:00",
        "lastTriggerTime": "2024-04-11T15:12:20.000+08:00",
        "triggerCronExp": "0 0 19 ? * 2,6,1",
        "restoreCronExp": "0 0 21 ? * 2,6,1",
        "selectedFieldNames": [
          "custFeature1",
          "promotionText"
        ],
        "triggerType": 1,
        "applyJobId": null,

```

```
"restoreJobId": null,
"periodType": 1,
"periodValue": [
  2,
  6,
  1
],
"periodTriggerTime": "19:00:00",
"periodRestoreTime": "21:00:00",
"periodTimes": [
  "19:00:00",
  "21:00:00"
],
"templateAttrName": "default",
"templateCategoryName": "default",
"selectFieldNameNum": [
  5,
  4
],
"createdTimeStr": "2024-04-11 14:30:12",
"lastTriggerTimeStr": "2024-04-11 15:12:20",
"startDateStr": "2024-04-10 00:00:00",
"endDateStr": "2024-04-30 23:59:59"
},
"totalElements": 1
},
"translate": true,
"originalMessage": null
}
```

Example 2

```
{
  "success": false,
  "code": 19017,
  "message": "At least one of external store ID and store ID must not be empty",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

8.4 Query Strategy Details

API Definition

URL	http://[server address]/zk/strategy/get/{id}
-----	--

Example	http://esl.zkong.com/zk/strategy/get/457
Request Method	GET
Content Type	application/json;charset=utf-8
Description	Query promotion strategy details.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Strategy ID

Request Example

Request URL:
<http://esl.zkong.com/zk/strategy/get/457>

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 102)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": {
    "id": 457,
    "name": "Limited-time Discount Strategy - New",
    "agencyId": 1684844103168,
    "merchantId": 1684844150922,
  }
}
```

```
"storeId": 1684845118494,
"code": "2240457",
"templateAttrCategory": "default",
"templateAttr": "default",
"startDate": "2024-04-10T00:00:00.000+08:00",
"endDate": "2024-04-30T23:59:59.000+08:00",
"status": 1,
"creator": "yintai",
"createdTime": "2024-04-11T14:30:12.000+08:00",
"updateTime": "2024-04-11T16:54:19.000+08:00",
"lastTriggerTime": "2024-04-11T15:12:20.000+08:00",
"triggerCronExp": "0 0 19 ? * 2,6,1",
"restoreCronExp": "0 0 21 ? * 2,6,1",
"strategyItems": [
{
  "id": 7731,
  "itemTitle": "cola550ml",
  "agencyId": 1684844103168,
  "merchantId": 1684844150922,
  "storeId": 1684845118494,
  "isFlag": 1,
  "price": "3.5",
  "memberPrice": "2.5",
  "unit": null,
  "classLevel": "level",
  "productArea": "China",
  "originalPrice": "4.35",
  "promotionText": "Limited-time Special Offer",
  "custFeature1": "Until the end of the month",
  "custFeature2": null,
  "custFeature3": null,
  "custFeature4": null,
  "custFeature5": null,
  "custFeature6": null,
  "custFeature7": null,
  "custFeature8": null,
  "custFeature9": null,
  "custFeature10": null,
  "custFeature11": null,
  "custFeature12": null,
  "custFeature13": null,
  "custFeature14": null,
  "custFeature15": null,
  "item": {
    "id": 11184969707,
    "type": 1,
    "price": 4.35,
    "spec": null,
    "productCode": "1265",
    "barCode": "6948939673411",
    "agencyId": 1684844103168,
```

```
"merchantId": 1684844150922,
"storeId": 0,
"originalPrice": 4.35,
"memberPrice": 2.99,
"productArea": "China",
"classLevel": "level",
"attrCategory": "default",
"attrName": "default",
"sourceCode": null,
"proStartTime": "2018-10-10 10:00:00",
"proEndTime": "2018-10-10 10:00:00",
"unit": null,
"productSku": "123",
"createTime": "2023-10-27 17:14:46",
"itemTitle": "cola550ml",
"shortTitle": "cola",
"qrCode": null,
"promotionText": "discount",
"custFeature1": null,
"custFeature2": null,
"custFeature3": null,
"custFeature4": null,
"custFeature5": null,
"custFeature6": null,
"custFeature7": null,
"custFeature8": null,
"custFeature9": null,
"custFeature10": null,
"custFeature11": null,
"custFeature12": null,
"custFeature13": null,
"custFeature14": null,
"custFeature15": null,
"custFeature16": null,
"custFeature17": null,
"custFeature18": null,
"custFeature19": null,
"custFeature20": null,
"custFeature21": null,
"custFeature22": null,
"custFeature23": null,
"custFeature24": null,
"custFeature25": null,
"custFeature26": null,
"custFeature27": null,
"custFeature28": null,
"custFeature29": null,
"custFeature30": null,
"custFeature31": null,
"custFeature32": null,
"custFeature33": null,
```

```
"custFeature34": null,
"custFeature35": null,
"custFeature36": null,
"custFeature37": null,
"custFeature38": null,
"custFeature39": null,
"custFeature40": null,
"custFeature41": null,
"custFeature42": null,
"custFeature43": null,
"custFeature44": null,
"custFeature45": null,
"custFeature46": null,
"custFeature47": null,
"custFeature48": null,
"custFeature49": null,
"custFeature50": null,
"updateTime": "2023-10-27 17:14:46",
"stock1": null,
"stock2": null,
"stock3": null,
"nfcUrl": null,
"storeName": null,
"label": "",
"operationUserId": 374
}
},
"selectedFieldNames": [
"custFeature1",
"promotionText"
],
"triggerType": 1,
"applyJobId": null,
"restoreJobId": null,
"periodType": 1,
"periodValue": [
2,
6,
1
],
"periodTriggerTime": "19:00:00",
"periodRestoreTime": "21:00:00",
"periodTimes": [
"19:00:00",
"21:00:00"
],
"templateAttrName": "default",
"templateCategoryName": "default",
"selectFieldNameNum": [
5,
```

```
4
],
"createdTimeStr": "2024-04-11 14:30:12",
"lastTriggerTimeStr": "2024-04-11 15:12:20",
"startDateStr": "2024-04-10 00:00:00",
"endDateStr": "2024-04-30 23:59:59"
},
"translate": true,
"originalMessage": null
}
```

Example 2

```
{
"success": false,
"code":21015,
"message": "Strategy configuration information does not exist",
"data": null,
"translate": true,
"originalMessage": null
}
```

8.5 Delete promotion strategy.

API Definition

URL	http://[server address]/zk/strategy/delete/{id}
Example	http://esl.zkong.com/zk/strategy/delete/457
Request Method	DELETE
Content Type	application/json;charset=utf-8
Description	Delete promotion strategy.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Strategy ID

Request Example

Request URL:

<http://esl.zkong.com/zk/strategy/delete/457>

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 99 102)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 21015,
  "message": "Strategy configuration information does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

8.6 Enable promotion strategy.

API Definition

URL	http://[server address]/zk/strategy/enable/{id}
Example	http://esl.zkong.com/zk/strategy/enable/457
Request Method	PUT

Content Type	application/json;charset=utf-8
Description	Enable promotion strategy.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Strategy ID

Request Example

Request URL:
<http://esl.zkong.com/zk/strategy/enable/457>

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details
message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 97 102)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 21015,
  "message": "Strategy configuration information does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

8.7 Disable promotion strategy.

API Definition

URL	http://[server address]/zk/strategy/disable/{id}
Example	http://esl.zkong.com/zk/strategy/disable/457
Request Method	PUT
Content Type	application/json;charset=utf-8
Description	Disable promotion strategy.

Request Parameters

Parameter Name	Parameter Type	Data Type	Required	Notes
Authorization	header	String	true	Authorization token
id	path	Integer	true	Strategy ID

Request Example

Request URL:
<http://esl.zkong.com/zk/strategy/disable/457>

Response Parameters

Parameter Name	Description	Data Type	Notes
success	Request Successful	Boolean	true/false
code	Status Code	Integer	See Appendix 1 for details

message	Message	String	Message
data	Response Data	Json	

Response Example (related error code no.: 102)

```
{
  "success": true,
  "code": 10000,
  "message": "Operation successful",
  "data": null,
  "translate": true,
  "originalMessage": "success"
}
```

Example 2

```
{
  "success": false,
  "code": 21015,
  "message": "Strategy configuration information does not exist",
  "data": null,
  "translate": true,
  "originalMessage": null
}
```

Appendix 1: API Error Code Reference

No.	Error Code	Message	Notes
1	11111	Operation Failed	General failure error code
2	14018	Username does not exist	Please check carefully before logging in
3	10030	You do not have permission to access this	No menu permission or store permission
4	10031	This user has been locked due to too many failed login attempts	Please try again later (locked for 3 hours)
5	14043	Account or password is incorrect	Please check carefully before logging in
6	14056	Your password has expired	Please contact the administrator to change the password
7	14041	This merchant account status is unavailable	Please contact the administrator to enable the merchant
8	14038	This account is disabled and cannot log in	Please contact the administrator to enable the

			account
9	10046	The merchant of the current user is not authorized	Please log in to the web portal and upload the authorization file.
10	19001	Product list is empty	Please check and then import
11	19014	Product list length is greater than 20,000	The single-import quantity exceeds the limit
12	19016	External store ID cannot be empty	Required when using external store ID
13	19015	No store matches the external store ID	Please check and then import
14	19018	Store does not exist or is disabled	Please confirm the store information before importing
15	19019	Agency ID does not match	Please confirm that the agency ID information is correct
16	19027	Merchant ID does not match	Please confirm that the merchant ID information is correct
17	17054	The number of product imports today exceeds the limit	The number of products imported in a day cannot exceed 10 times the license limit
18	17055	The total number of merchant products exceeds the limit	The total number of imported products exceeds the license limit
19	10001	The system is busy. Please try again later	Current service exception
20	10047	The current API is busy. Please try again later	Parallel imports are not allowed for the same merchant
21	17002	successful import of merchandise	All products imported successfully
22	17050	successful import part of merchandise	Some products imported successfully (product error information will be returned)
23	17051	unsuccessful import of merchandise	All product imports failed (product error information will be returned)
24	19002	Product list length is greater than 500	This API limits the product quantity to 500
25	17006	Product code format is incorrect	Empty values and certain special characters are not supported
26	17052	item used by store	The product is in use and cannot be deleted
27	17004	Product does not exist	Please check whether the barcode is correct
28	10041	Image file cannot be empty	Please check the image file parameters
29	17009	Image file format error	Image file format error
30	10037	Image format is incorrect	Please check the image format
31	10038	Image size is too large	Image size exceeds the API limit

32	10042	Image dimensions do not match	Actual image dimensions do not match the parameters
33	10049	Image processing method is incorrect	Please check the image parsing parameters
34	10045	ESL parameter error	Please compare against the parameter ranges supported by the document
35	17018	The product already exists or the product code is already in use. Do not add it again	Please check the product data
36	17053	Image parameters do not match the ESL model data	Please check the input parameters
37	10016	Upload failed	File upload failed
38	19003	Store ID does not match	Please check the store parameters
39	10039	Image index can only be 1-10	Please check the image index parameter
40	17039	The barcode is already used by an image product. Import failed	Please check the product code
41	10009	param invalid	Invalid parameters (required parameters are missing or the format is incorrect)
42	13040	Binding relationship does not exist	Please check the ESL binding information
43	13023	The ESL does not belong to this store	Please confirm the ESL ownership
44	15045	ESL barcode format is incorrect	Please check the ESL barcode parameter
45	19017	At least one of external store ID and store ID must not be empty	Please check the store ID parameter
46	15069	Device not found. Please contact the agency.	Please confirm the device information
47	15035	The new device has not reported its status yet. Please try again later	Please wait for the device to report or specify an AP for image refresh
48	13037	The API batch unbind quantity cannot exceed 2,000	
49	14031	Insufficient permissions	No device usage permission
50	13030	some esl are unbound	Some ESLs failed to unbind
51	13035	The API batch bind quantity cannot exceed 2,000	
52	13029	The number of bound ESLs under the merchant exceeds the limit	Please contact sales to update the license limit
53	17038	Duplicate products exist	Please check the product information
54	17037	Image products cannot be used	
55	13020	The ESL does not belong to this merchant	Please verify the ESL ownership information
56	13027	The template size or color does not match the ESL.	Please check the template information

57	13026	The binding relationship between this product and ESL already exists	Please check the binding information
58	15016	The entered ESL code is incorrect or the queried ESL does not exist	Please check the ESL barcode information
59	13033	This ESL does not support inversion	Please confirm the ESL model and function
60	13022	The ESL is unbound	Please confirm the ESL status
61	14040	The queried store does not exist	Please confirm the store information
62	13039	The number of ESLs for LED flashing cannot exceed 1,000	Please reduce the length of the transmitted list
63	40005	ESL does not exist	Please check the ESL information
64	51103003	Flashing brightness is abnormal	Only 1-100 is supported
65	19006	Store ID cannot be empty	Please check the store ID parameter
66	15015	The ESL information you queried does not exist	Please check the ESL barcode information
67	21002	Cron expression error	Please check the parameter format
68	15060	AP device not found	Please check the device information
69	15059	AP already exists	Do not add it again
70	15058	ESL model does not exist	Please check the ESL information
71	12042	The template file format is incorrect	Please check the template file
72	12043	Your current operation content exceeds the maximum limit	Please confirm and submit again
73	12040	The template file is invalid; parsed content is empty	Please check the template file
74	12047	The uploaded product template type is incorrect	Please check the template type parameter
75	12041	Template content is invalid; fields are missing	
76	15075	ESL model is missing	Please check the ESL model parameter
77	12028	This template is enabled and bound to ESLs	Please confirm the template status
78	12036	This template is already enabled	Do not enable it again
79	12037	The number of templates to copy cannot exceed 100	Please reduce the number of templates to copy
80	12019	The template attribute category already exists	Do not add it again
81	10020	Name cannot be empty	Please enter the name information
82	12027	At least one template category-template attribute must exist under this	At least one template category/attribute must be retained

		merchant.	
83	12020	Attributes under this category are used by products or templates and cannot be deleted.	Please confirm before deleting
84	12025	Template category does not exist	Please confirm the template category information
85	12016	Template attribute already exists	Please confirm the template attribute information
86	12017	This attribute is used by a product or template and cannot be deleted.	Please confirm before deleting
87	12045	No template selected	Please check the template information parameters
88	12044	Template content is incomplete; information is missing	Please check the template information parameters
89	12023	Some ESLs are using this template. Disable this template?	Please confirm whether to force disable
90	14054	The external store ID already exists under this merchant	Please check the external store ID parameter
91	14009	Add failed; the store already exists	Please check whether the store information already exists
92	13031	The store quantity limit has been exceeded. New stores cannot be added. Please contact the agency	Please contact the agency to modify the authorization
93	14036	Store name already exists	Please check whether the store name already exists
94	52023003	LCD device does not exist	Please check the device DUID
95	52023006	This device has been bound by another user	Please confirm device ownership
96	21001	The strategy validity period must be at least one day	Please confirm whether the strategy validity period is valid
97	21004	Strategy conflict exception	Please confirm whether the current strategy and an existing promotion strategy contain the same products and have overlapping trigger time periods
98	21006	Promotion strategies currently support only always-triggered and fixed-cycle trigger methods	Please check the triggerType value
99	21008	A promotion strategy cannot be deleted while it is in use	Please disable the strategy before deleting it
100	21009	Basic information cannot be updated while the promotion strategy is in use	Please disable the strategy before modifying its basic information
101	21013	The current strategy name already exists	Please confirm whether it duplicates an existing

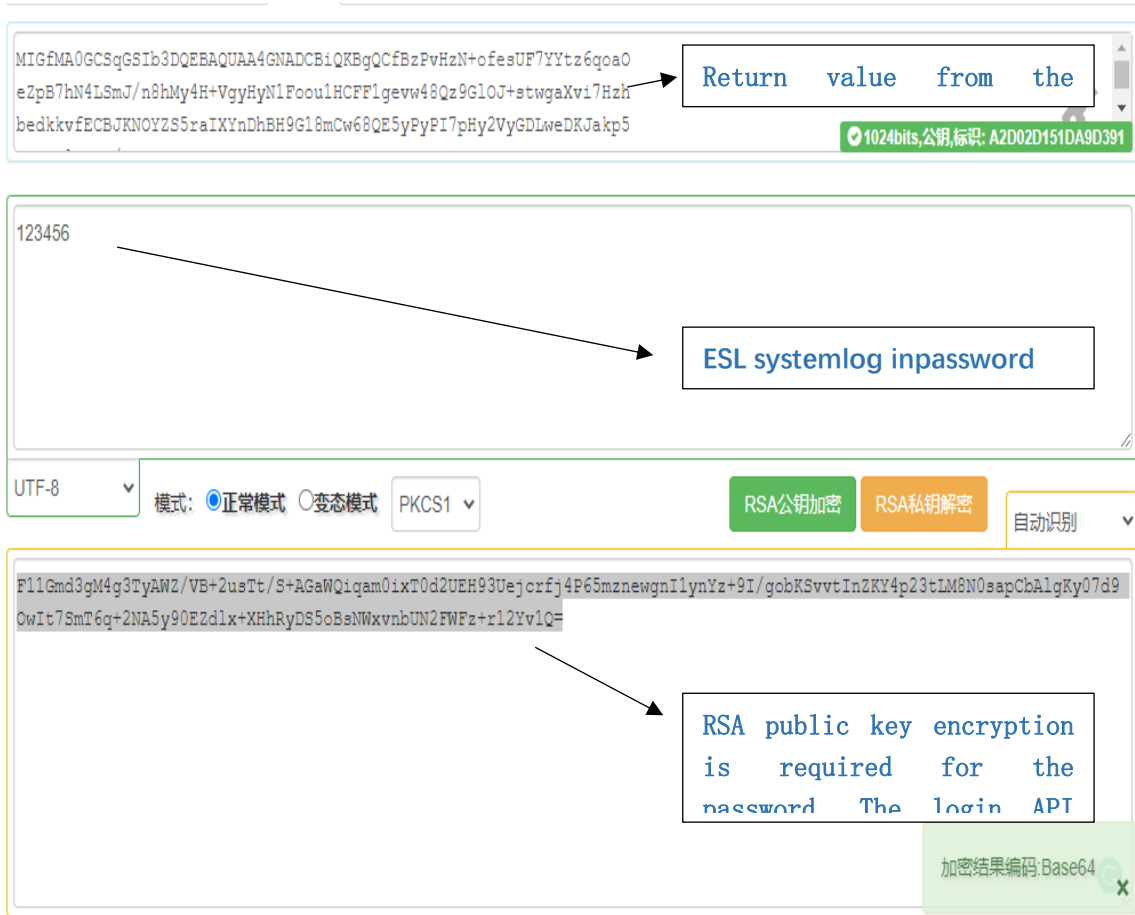
			strategy name
102	21015	Strategy configuration information does not exist	Please confirm whether the strategy ID exists

Appendix 2: RSA Public Key Encryption Example

Sample Code (Java)

```
public static String encryptByPublicKey(String publicKeyText, String password) throws Exception {
    X509EncodedKeySpec x509EncodedKeySpec2
    = new X509EncodedKeySpec(Base64.decodeBase64(publicKeyText));
    KeyFactory keyFactory = KeyFactory.getInstance("RSA");
    PublicKey publicKey = keyFactory.generatePublic(x509EncodedKeySpec2);
    Cipher cipher = Cipher.getInstance("RSA");
    cipher.init(Cipher.ENCRYPT_MODE, publicKey);
    byte[] result = cipher.doFinal(password.getBytes());
    return Base64.encodeBase64String(result);
}
```

Reference URL: <https://the-x.cn/Cryptography/Rsa.aspx>



Appendix 3: LCD Operation Command Examples

Operation Type (type)	Description	Notes
1	Start/Pause Playback	
2	Power Off	
3	Restart System	
4	Restart Player	
5	Adjust Volume	value indicates the volume
6	Adjust Brightness	value indicates the brightness
7	Power On/Off Configuration	value indicates the configuration content

8	Report Information	Request the device to actively report information
9	Delete Device	Clear Playback Content
10	Publish Playback Content	The message does not contain a type; the ACK reply type for this message is 10
11	Issue Boot Logo	value is the logo URL
12	Switch Wi-Fi Network Mode	If switching to scheduled network connection, scheduled settings will be issued. If empty, switches to always-connected mode.
13	Show/Hide QR Code	0: hide, 1: show
14	App Update Command	After receiving the command, call the API to download the resource
15	System Update Command	After receiving the command, call the API to download the resource
16	Issue App Startup Logo	value is the logo URL
17	Enable the "stop playback timeout" function	
18	QR code/barcode switch	1: QR code, 2: barcode
19	Issue Network Configuration Settings	Issue up to 10 groups of SSID/password and replace the original settings. During NFC network configuration, if there are fewer than 10 current network settings, append the new one; otherwise replace the 10th setting. (this setting does not affect the factory default configuration)
20	Scheduled Network Connection Settings	Value has 6 attributes: networkState: 0off 1enabled interval: scheduledcycle(Unit/seconds) timeout: connection timeout (unit: seconds) reconnectNum: failed reconnect attempts reconnectSet: reconnect parameter settings firstConnectTime: first connection time after power-on See Example 5 below
21	Pull Publication Notification Command	After receiving it, request API 6.1 publication list
22	Format Device Storage	Format device storage after receiving the command

23	Report Logs	
24	AI Publication	
25	Playback Content Rotation	value is the rotation angle: 0/90/180/360
26	Log Recording Setting	0 - off, 1 - on
27	Merchant Time Zone Update	After receiving it, request API 6.4 to get the new merchant time zone

Example 1 (Start/Pause Playback)

```
{
  "type": 1, //Operation Type
  "value": "" //operationcontent/configurationinformation
}
```

Example 2 (Adjust Volume)

```
{
  "type": 5, //Operation Type
  "value": "40" //operationcontent
}
```

Example 3 (Power On/Off Configuration)

```
{
  "type": 7, //Operation Type
  "value": "{
    "startTime": 30, //configuration effective start time
    "endTime": 30, //configuration effective end time
    "switchTimeConfig": [
      {
        "type": 1, //type: 1 daily, 2 weekly, 3 fixed date (only one of each type)
        "switchTime": [
          {
            "onTime": "00: 00", //power-on time
            "offTime": "05: 00" //power offtime
          },
          {
            "onTime": "06: 00",
            "offTime": "10: 00"
          }
        ]
      }
    ]
  }
```

```
{
  "type": 2,
  "weekday": [ //days of the week
    1,
    2,
    3
  ],
  "switchTime": [
    {
      "onTime": "00: 00",
      "offTime": "05: 00"
    },
    {
      "onTime": "06: 00",
      "offTime": "10: 00"
    }
  ]
},
{
  "type": 3,
  "switchTime": [
    {
      "date": "2021-08-10", //fixed date
      "onTime": "00: 00",
      "offTime": "05: 00"
    },
    {
      "date": "2021-08-15",
      "onTime": "05: 00",
      "offTime": "10: 00"
    }
  ]
}
}
```